EE6108 Workshop Laboratory Manual

INSTRUCTOR KEY & FACILITATION GUIDE

Course Title: Practical Computer Networking Workshop
Target Audience: Engineering Students (General Engineering Background)
Simulation Engine: Cisco Packet Tracer

Module Code: EE6108

Estimated Workshop Duration: 3 Hours

Class Format: Team-Based Hands-on Engineering

Document Scope: Complete Solutions, IOS Scripts, & Assessment Rubrics

> **Instructor Facilitation Focus:** This manual provides complete answer keys, Cisco IOS configuration scripts, and real-time diagnostic checkpoints. Students frequently struggle with Layer 2 (MAC) vs. Layer 3 (IP) forwarding boundaries, the necessity of Default Gateways, Trunk encapsulation across switches, and standard ACL placement. Emphasize physical link convergence LEDs and verify reachability step-by-step.

Lab 1: Local Area Network Addressing & Connectivity Verification

1. Pedagogical Scenario & Teaching Objectives

Participants assemble a local area network (LAN) connecting two computers through a Layer 2 switch. Students test same-subnet and cross-subnet addressing to directly observe how subnet masks dictate local Layer 2 broadcast boundaries.

- Guide students through deploying endpoint PCs, a Catalyst 2960 switch, and automatic connection cabling.
- Reinforce static IPv4 host assignment via the Config tab.
- Demonstrate that hosts calculate broadcast domain boundaries locally via bitwise ANDing of the IP and mask.
- Train students to diagnose connectivity with `ping` and distinguish between physical link failures and protocol-level timeouts.

2. Master Addressing Key

Device	Interface	Initial Test IP	Target Corrected IP	Subnet Mask	Default Gateway	Verification State
PC0	FastEthernet0	192.168.1.1	192.168.1.1	255.255.255.0 (/24)	None (Direct L2)	Active Link (Green LED)
PC1	FastEthernet0	192.168.2.1	192.168.1.2	255.255.255.0 (/24)	None (Direct L2)	Active Link (Green LED)
Switch0	Fa0/1, Fa0/2	Unassigned	Unassigned	Layer 2 Transparent	N/A	VLAN 1 Default Forwarding

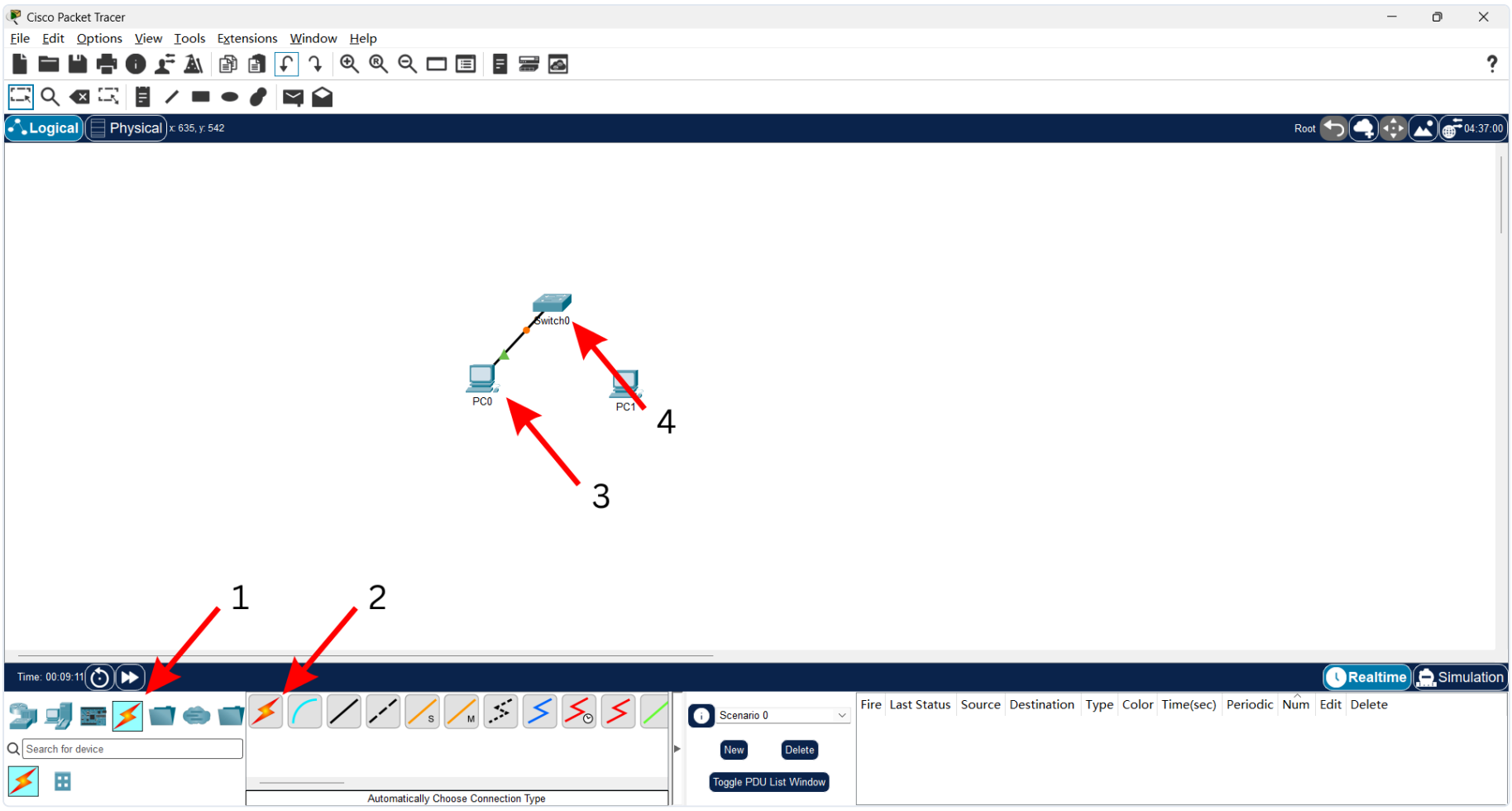
3. Step-by-Step Facilitation & Solution Checkpoints

Step 1: Physical Placement & Convergence

- Instruct students to place PC0, PC1, and Switch0 onto the workspace canvas.
- Use **Automatically Choose Connection Type** (lightning bolt icon) to cable the devices.

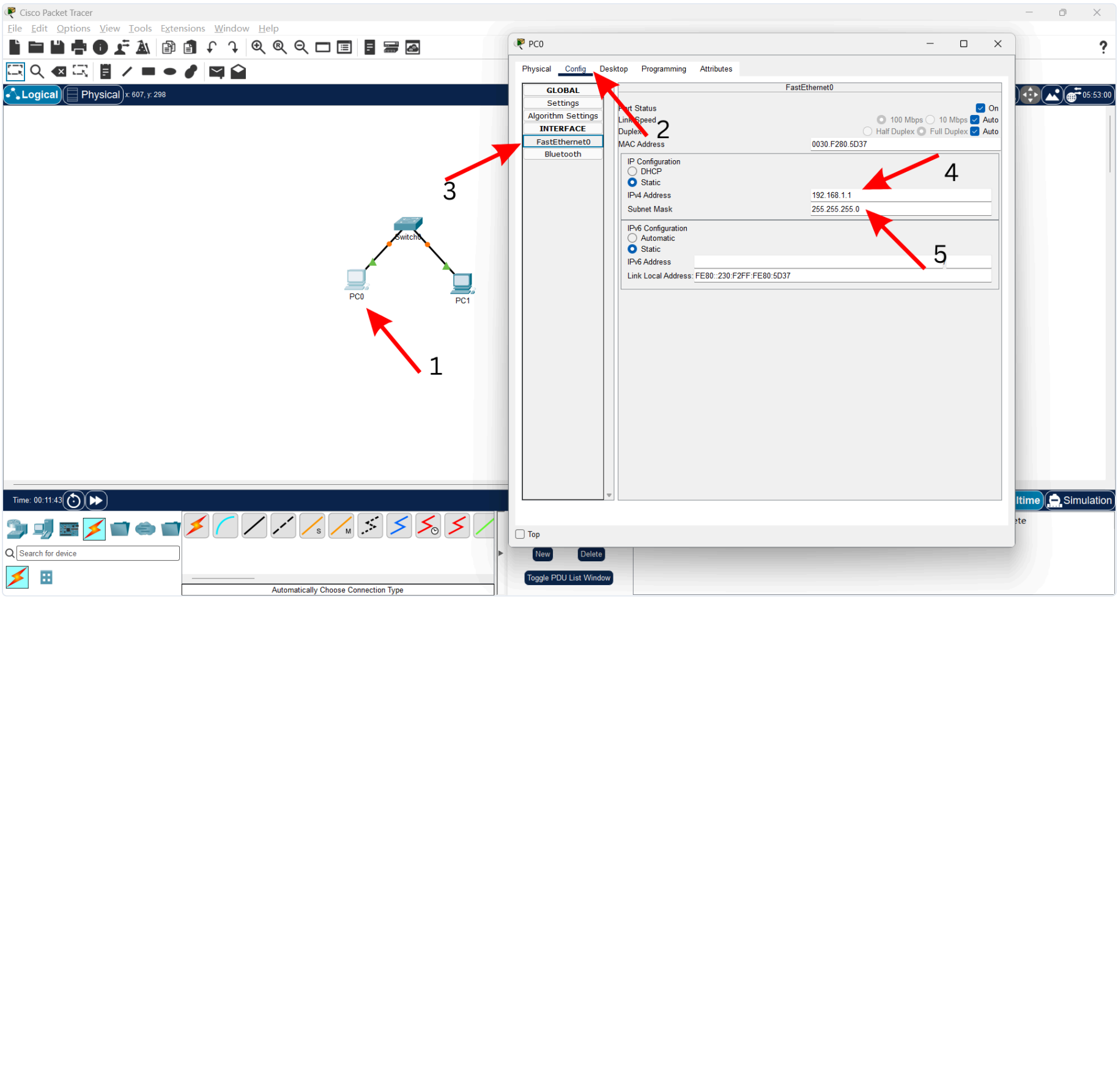
> **Instructor Checkpoint 1.1:** Switch ports display amber LEDs for roughly 30 seconds due to Spanning Tree Protocol (STP) listening and learning states. Instruct students to click "Fast Forward Time" in the bottom toolbar to accelerate transition to green.





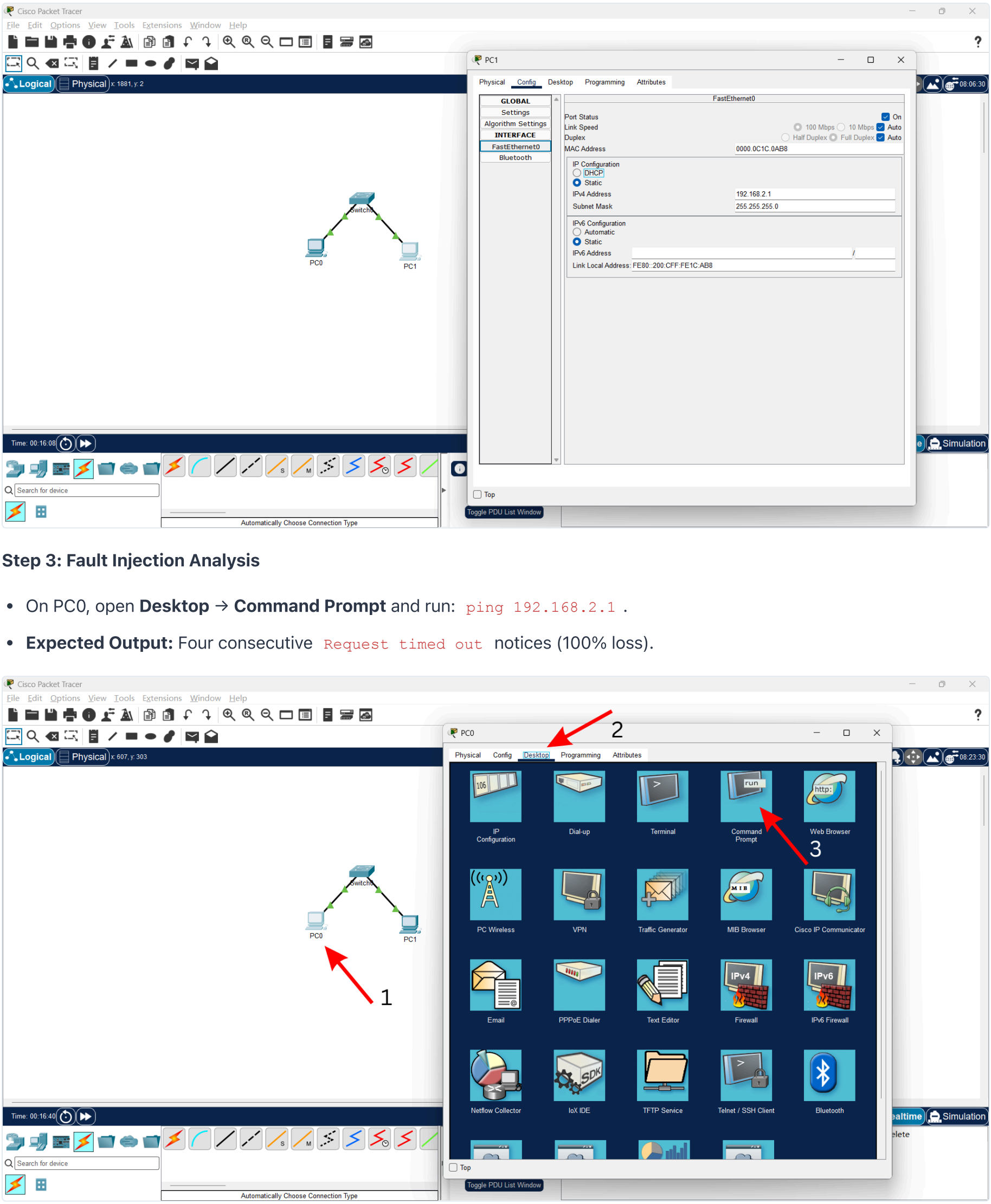
Step 2: Cross-Subnet Configuration (Controlled Failure Demonstration)

- PC0 Settings: Click **Config** → **FastEthernet0** → IP `192.168.1.1` , Mask `255.255.255.0` .
- PC1 Settings: Click **Config** → **FastEthernet0** → IP `192.168.2.1` , Mask `255.255.255.0` .



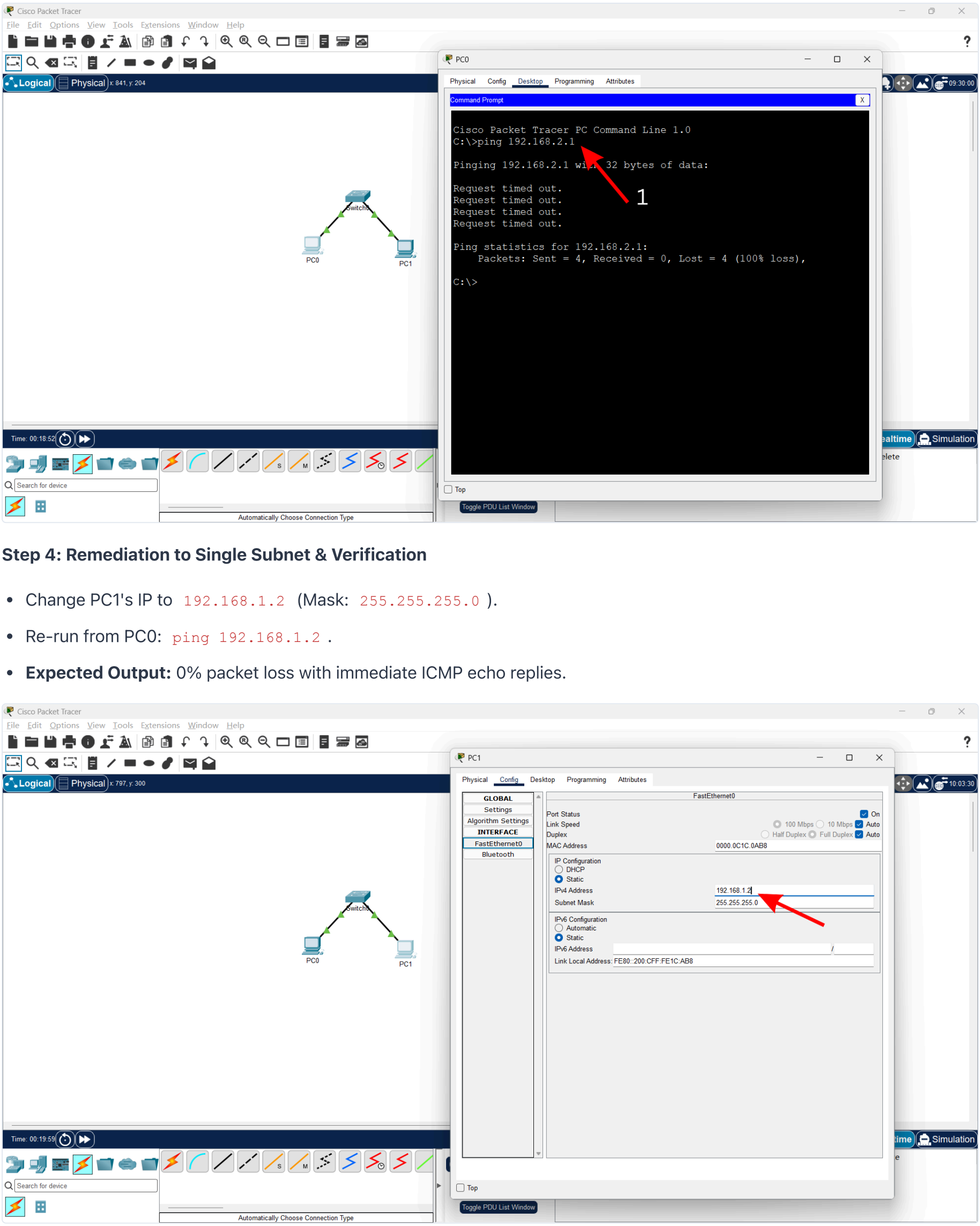
Step 3: Fault Injection Analysis

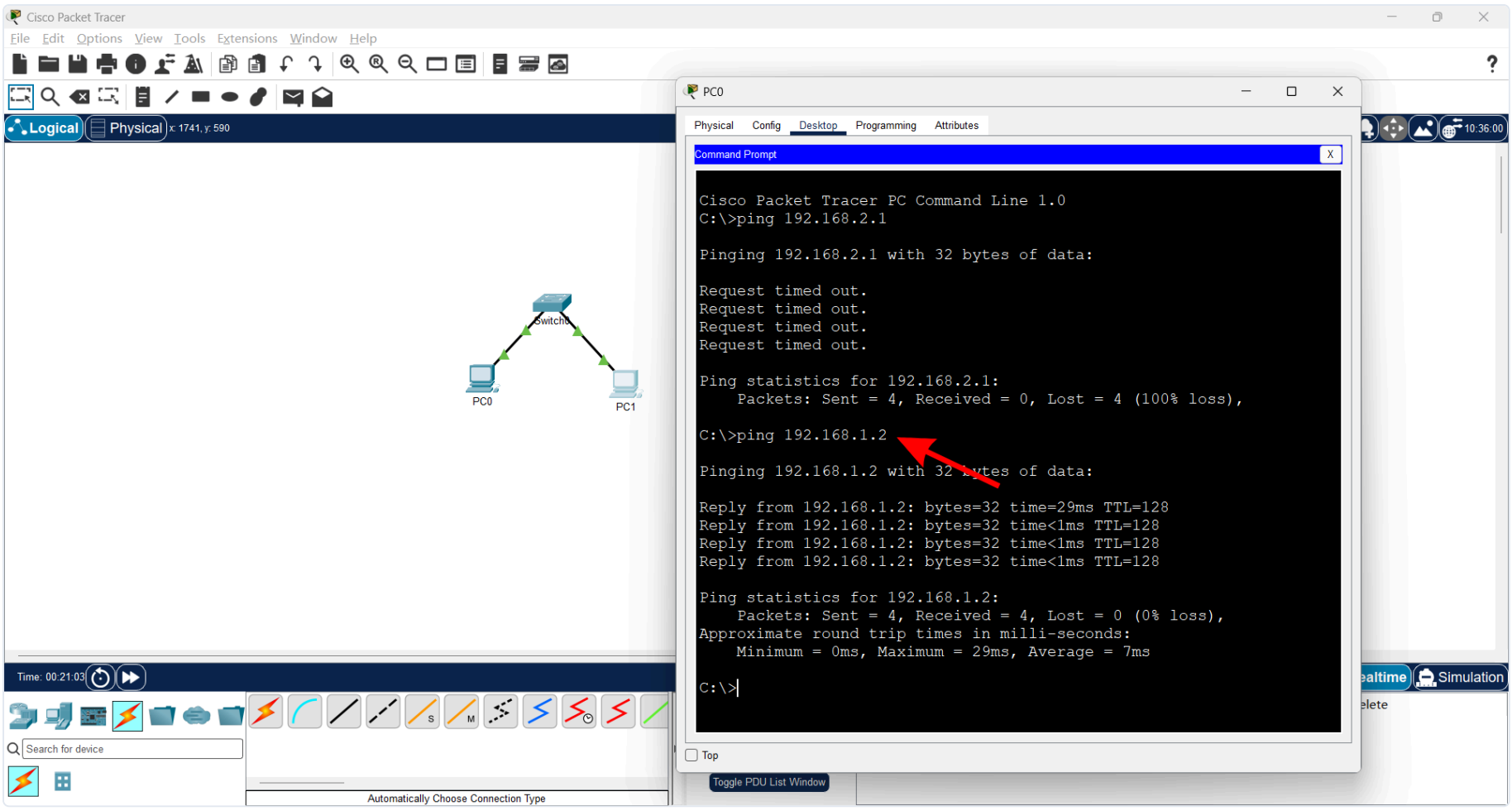
- On PC0, open **Desktop** → **Command Prompt** and run: `ping 192.168.2.1` .
- **Expected Output:** Four consecutive `Request timed out` notices (100% loss).



Step 4: Remediation to Single Subnet & Verification

- Change PC1's IP to `192.168.1.2` (Mask: `255.255.255.0`).
- Re-run from PC0: `ping 192.168.1.2`.
- **Expected Output:** 0% packet loss with immediate ICMP echo replies.





4. Lab 1 Question Solutions (Grading Guide)

Question 1: Subnet Identification & Host Routing Logic

Why was PC0 unable to communicate with 192.168.2.1 even though both devices were powered on and physically connected to the same switch?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Subnet Calculation (2 Marks):* Applying the subnet mask `255.255.255.0` (/24) isolates the first 24 bits as the Network ID. PC0 calculates its local network as `192.168.1.0`, while PC1 belongs to `192.168.2.0`.
> 2. *Off-Subnet Forwarding Rule (3 Marks):* Because the destination network does not match the local network, host PC0 determines that the target is off-subnet. TCP/IP architecture requires off-subnet packets to be forwarded through a Default Gateway. Because PC0 had no Default Gateway configured, it could not resolve an egress MAC address and discarded the packet locally, triggering timeouts.

Question 2: Layer 2 Switch Operations

Does the switch examine or modify the IP addresses of the packets passing through it? Which address format does the switch inspect?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Protocol Independence (2 Marks):* No. A standard Layer 2 switch operates strictly at the Data Link Layer and does not read, parse, or alter Layer 3 IPv4 headers.
> 2. *MAC Address Inspection (3 Marks):* The switch parses the 48-bit Media Access Control (MAC) address within the Ethernet frame header. It references its internal MAC Address Table (CAM table) to forward the frame directly out of the associated physical port.

Lab 2: Inter-Subnet Communication & Multi-Router Static Routing

1. Pedagogical Scenario & Teaching Objectives

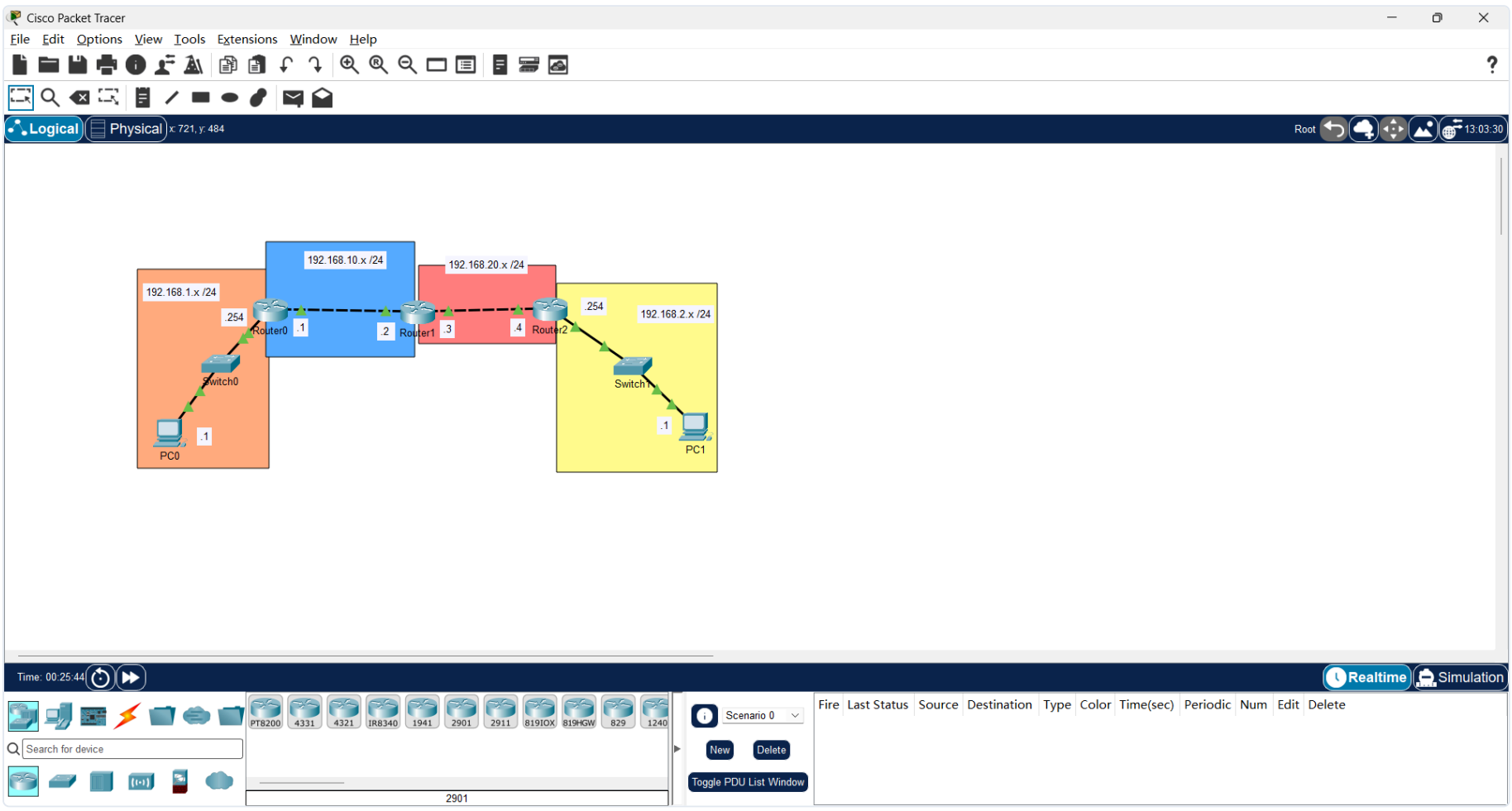
Connecting two distant workstations across three intermediary routers models a multi-hop transit network. Students configure Default Gateways on hosts and deploy Cisco IOS static routes (`ip route`) on each router to establish bidirectional reachability.

- Enforce the assignment of Default Gateways on endpoint computers.

- Train students on Cisco IOS interface hierarchy, IP assignment, and link activation with `no shutdown`.
- Deconstruct static route mechanics (`Destination Subnet + Mask + Next-Hop IP`) across multi-router chains.
- Verify end-to-end multi-hop connectivity using ICMP ping.

2. Master Topology & Addressing Matrix

> **Subnet Partition Architecture (/24 Class C Spaces):**
> - **Network A (Orange Zone):** `192.168.1.0/24` | Route0 LAN Port (.254)
> - **Transit Backbone 1 (Blue Zone):** `192.168.10.0/24` | Route0 ↔ Route1
> - **Transit Backbone 2 (Red Zone):** `192.168.20.0/24` | Route1 ↔ Route2
> - **Network B (Yellow Zone):** `192.168.2.0/24` | Route2 LAN Port (.254)

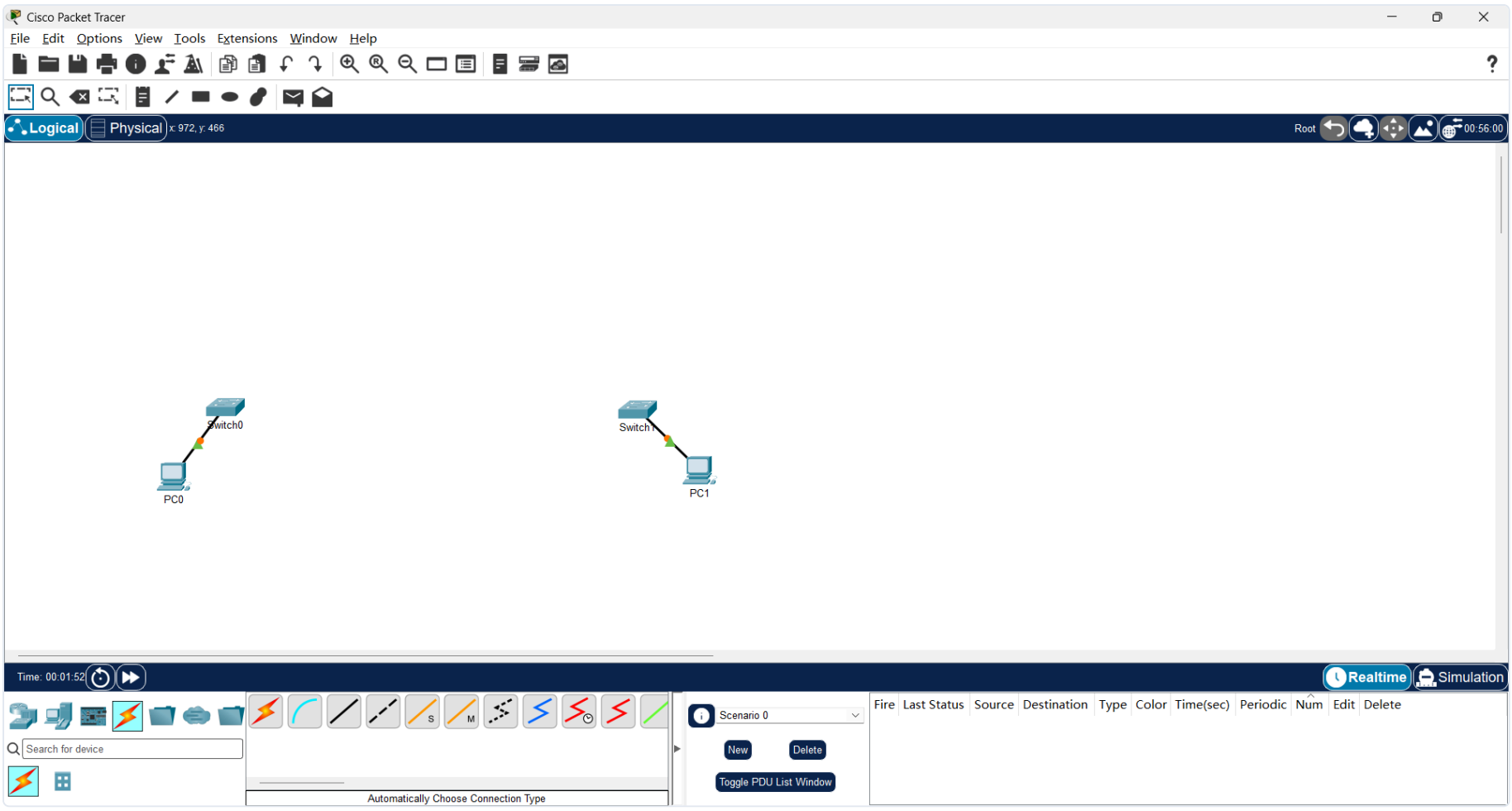


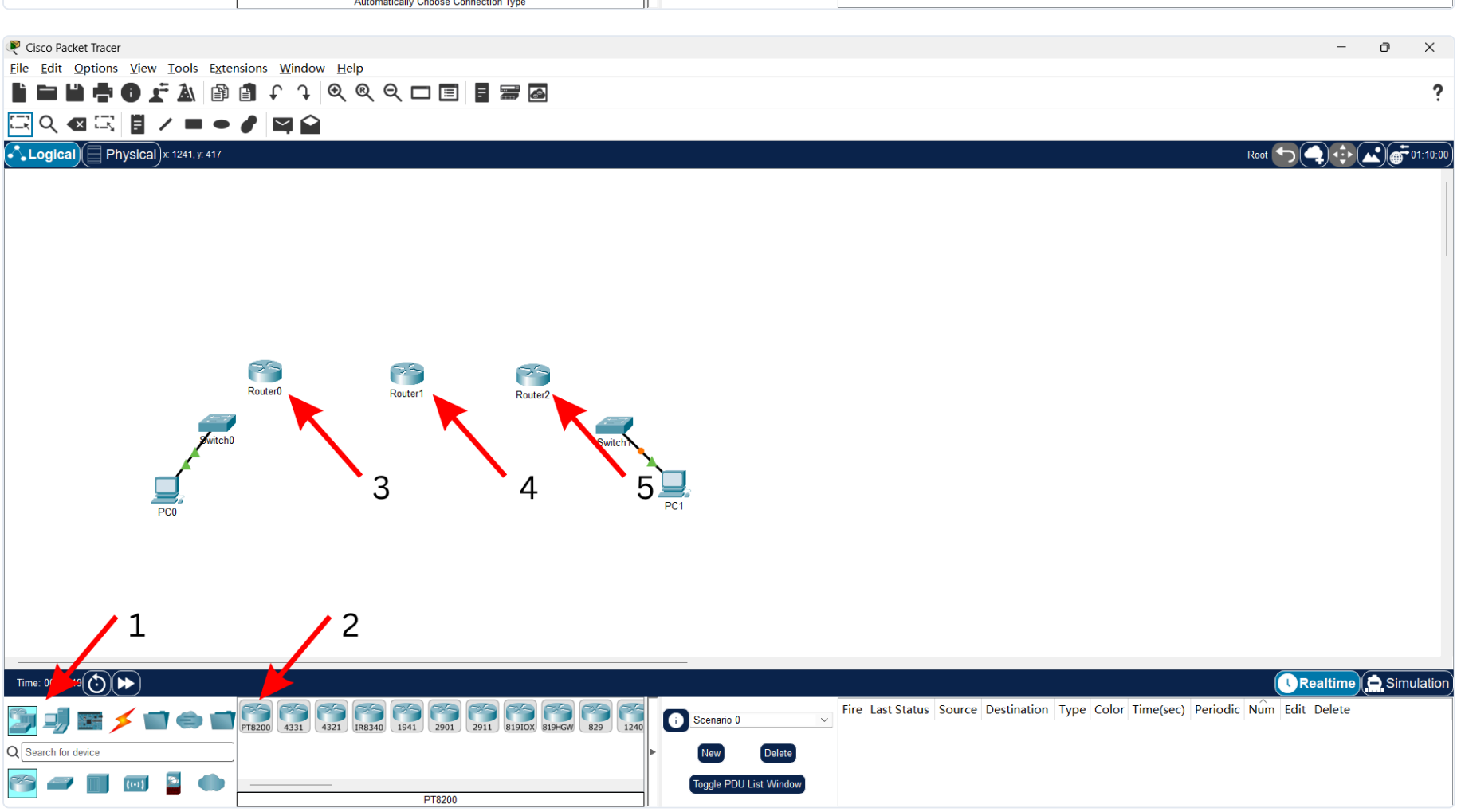
Device	Interface	IP Address	Subnet Mask	Default Gateway / Role	Expected Link State
PC0	FastEthernet0	192.168.1.1	255.255.255.0	**192.168.1.254**	Connected to Switch0
PC1	FastEthernet0	192.168.2.1	255.255.255.0	**192.168.2.254**	Connected to Switch1
Route0	Gig0/0/0	192.168.1.254	255.255.255.0	LAN Gateway for PC0	Up / Up (Manual)
	Gig0/0/1	192.168.10.1	255.255.255.0	WAN Link toward Route1	Up / Up (Manual)
Route1	Gig0/0/0	192.168.10.2	255.255.255.0	WAN Link toward Route0	Up / Up (Manual)
	Gig0/0/1	192.168.20.3	255.255.255.0	WAN Link toward Route2	Up / Up (Manual)
Route2	Gig0/0/0	192.168.20.4	255.255.255.0	WAN Link toward Route1	Up / Up (Manual)
	Gig0/0/1	192.168.2.254	255.255.255.0	LAN Gateway for PC1	Up / Up (Manual)

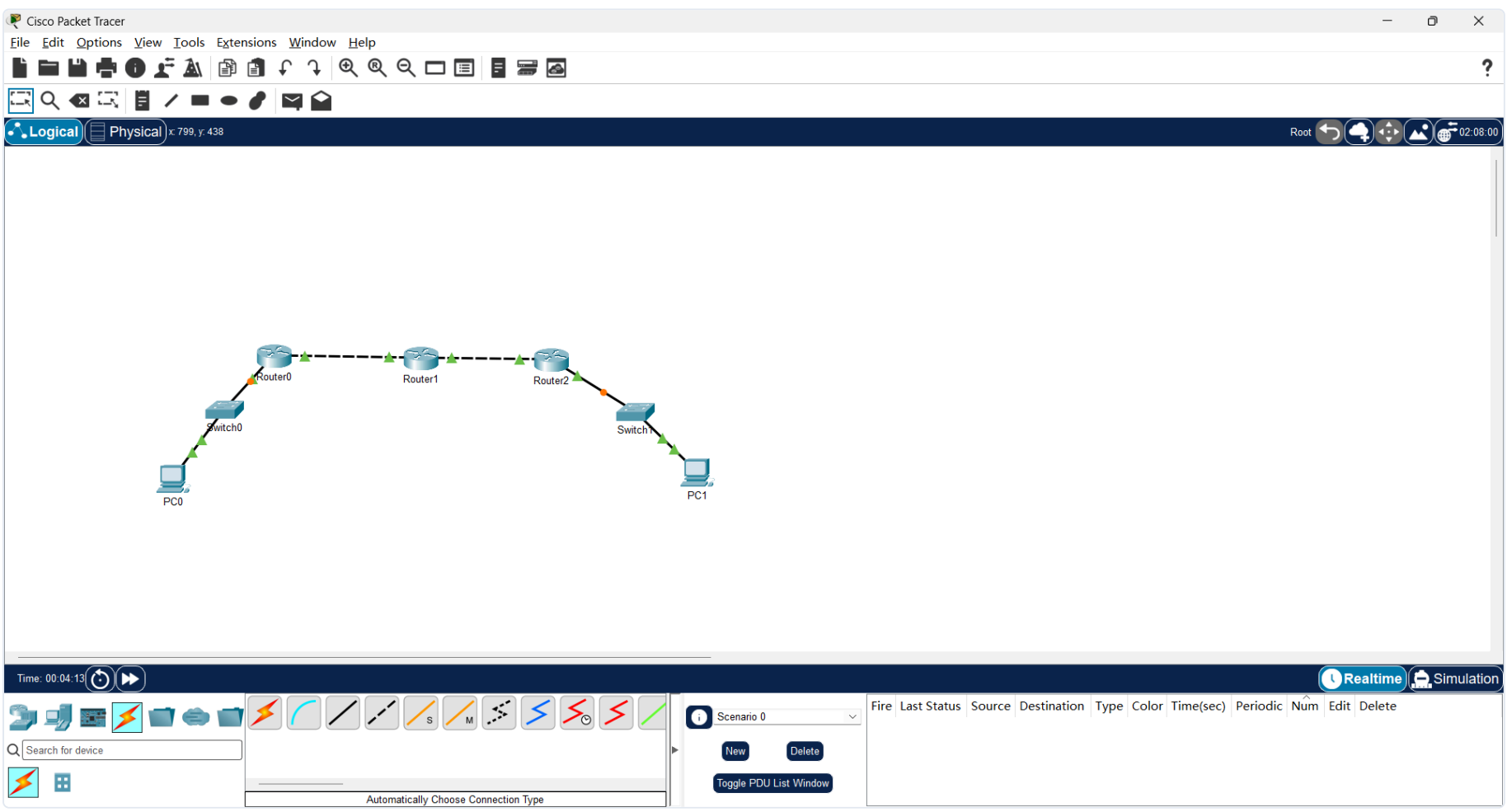
3. Step-by-Step IOS Commands & Verification Key

Step 1: Construct Topology and Cable

- Deploy 2 PCs, 2 switches (2960), and 3 routers. Connect linearly using connection cables.

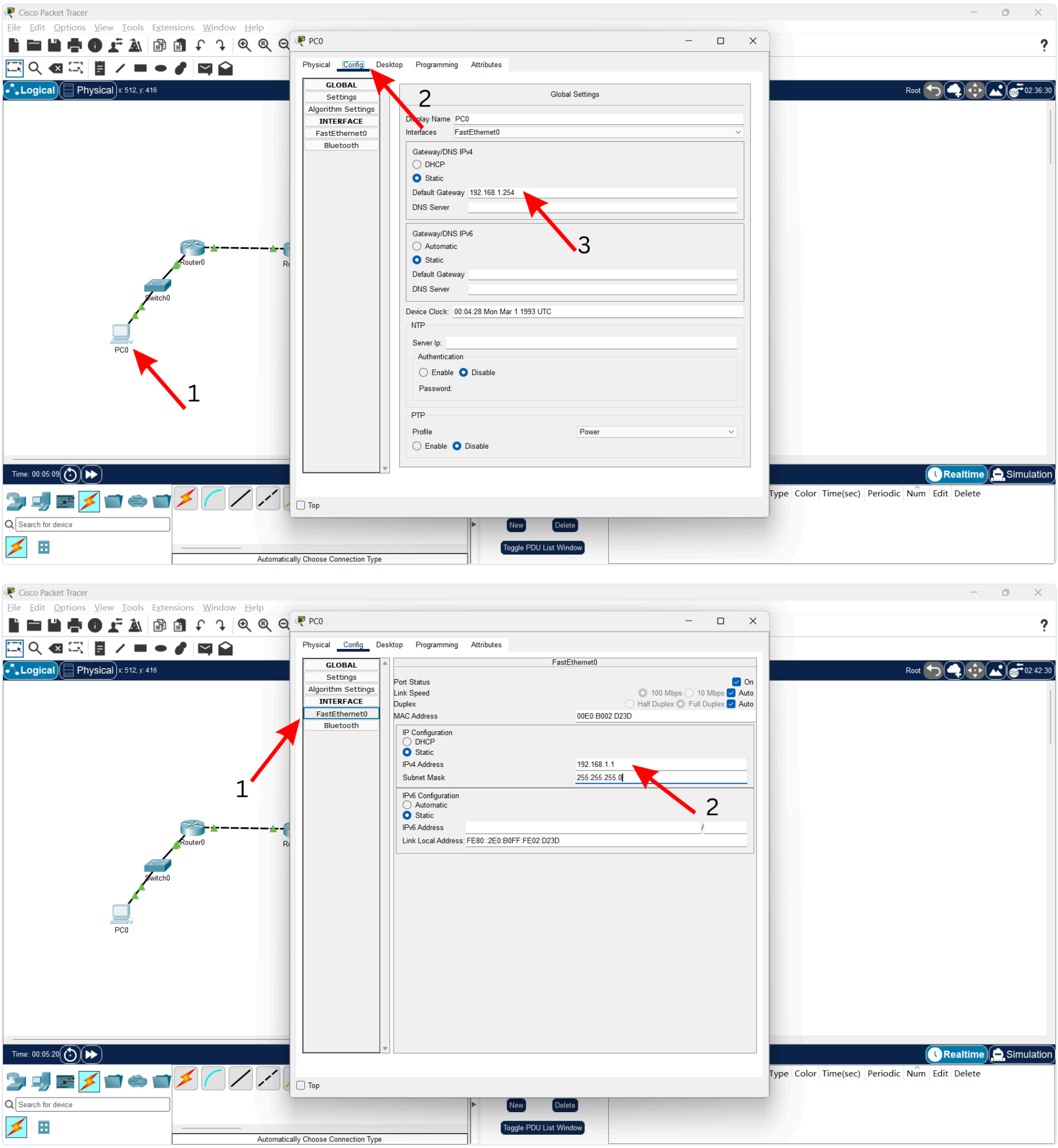

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 972, y: 466
Root 00:56:00
Switch0
Switch1
PC0
PC1
Time: 00:01:52
Realtime Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

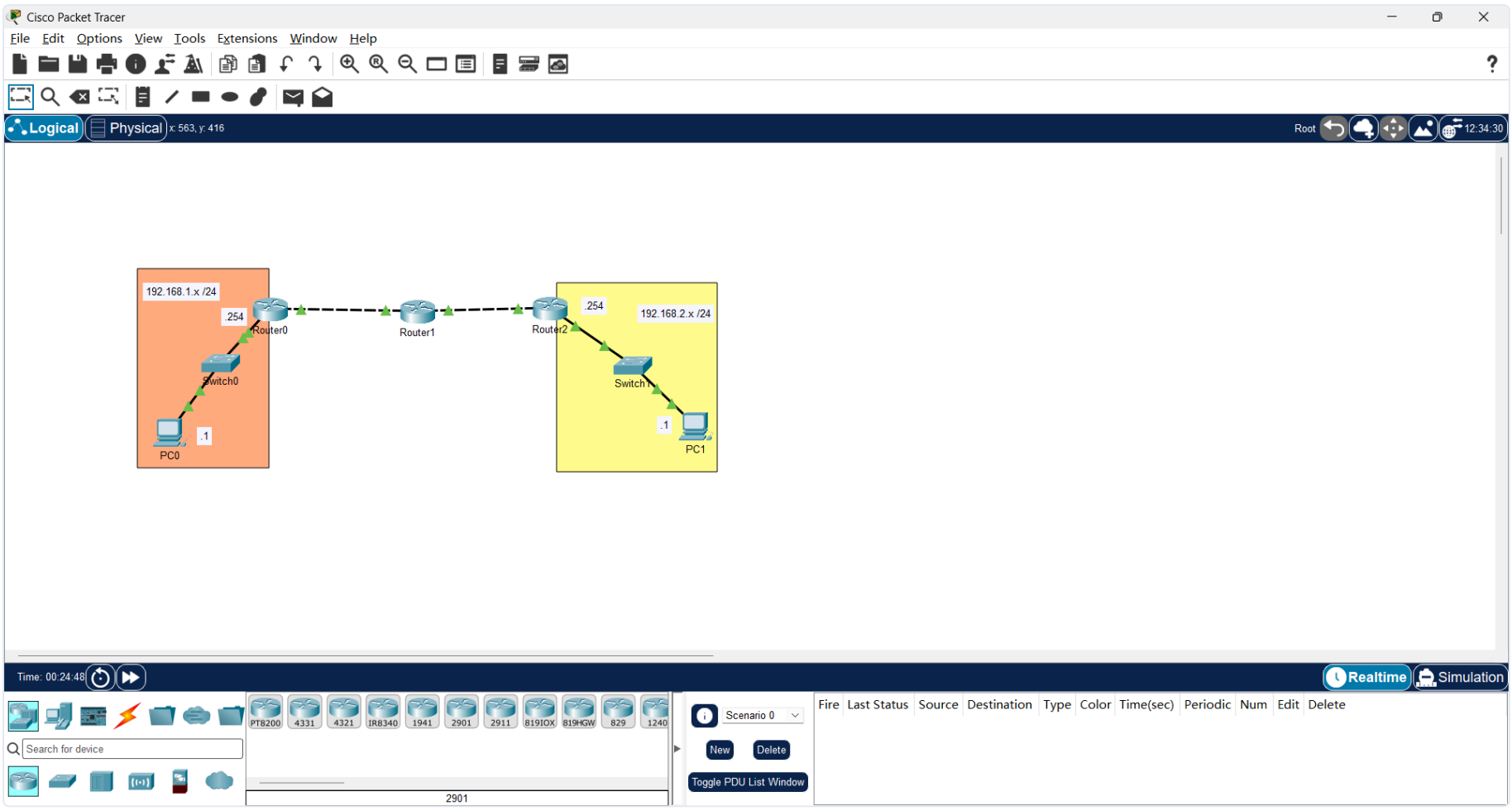

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1241, y: 417
Root 01:10:00
Router0
Router1
Router2
Switch0
Switch1
PC0
PC1
3
4
5
1
2
PT8200
PT8200
Time:
Realtime Simulation
Search for device
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 799, y: 438
Root 02:08:00
Router0
Router1
Router2
Switch0
Switch1
PC0
PC1
Time: 00:04:13
Realtime Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Step 2: Endpoint Host Configuration

- PC0: Click **Config** → **Global Settings** → Gateway `192.168.1.254` ; click **FastEthernet0** → IP `192.168.1.1` .
- PC1: Click **Config** → **Global Settings** → Gateway `192.168.2.254` ; click **FastEthernet0** → IP `192.168.2.1` .



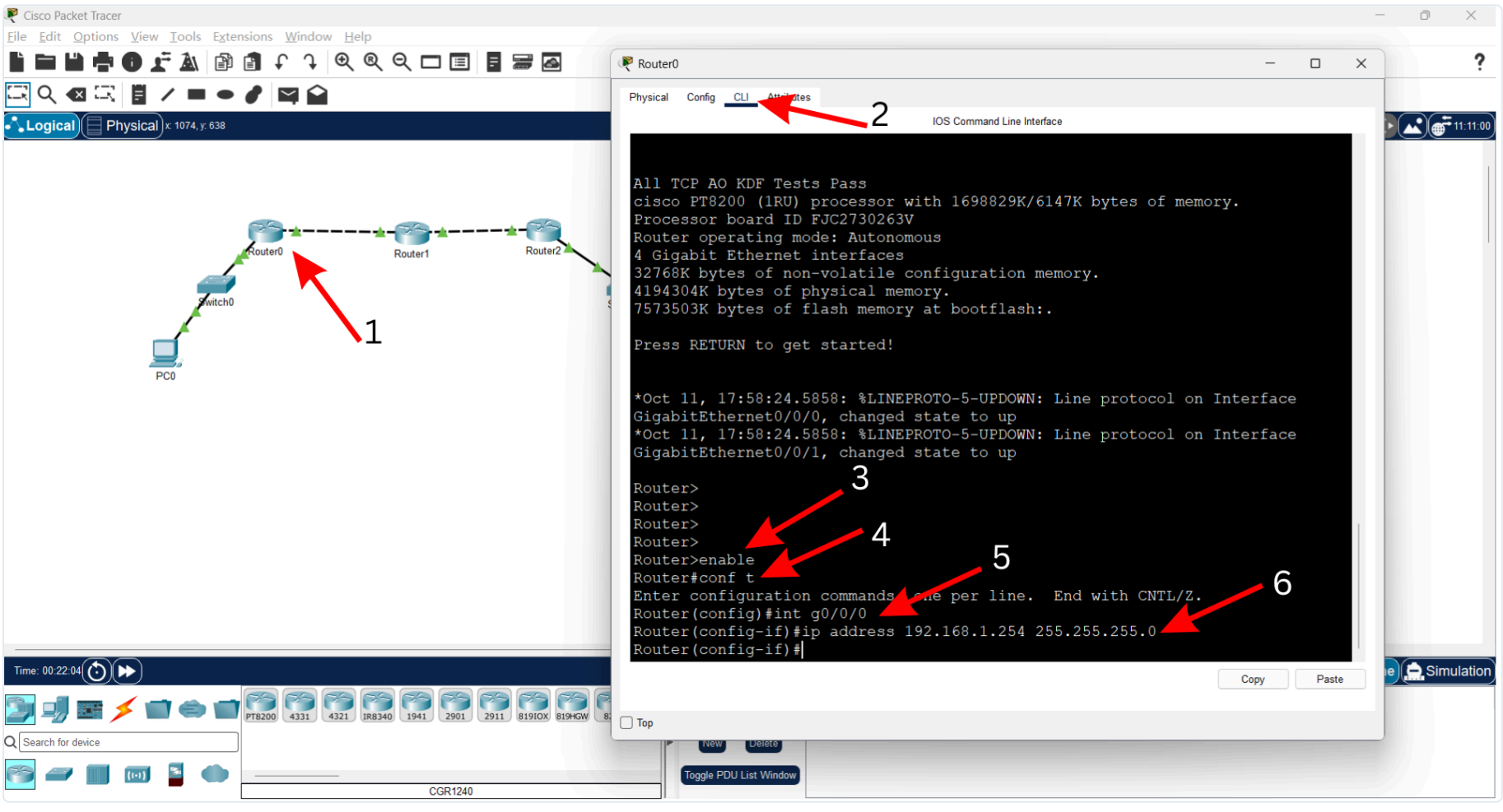


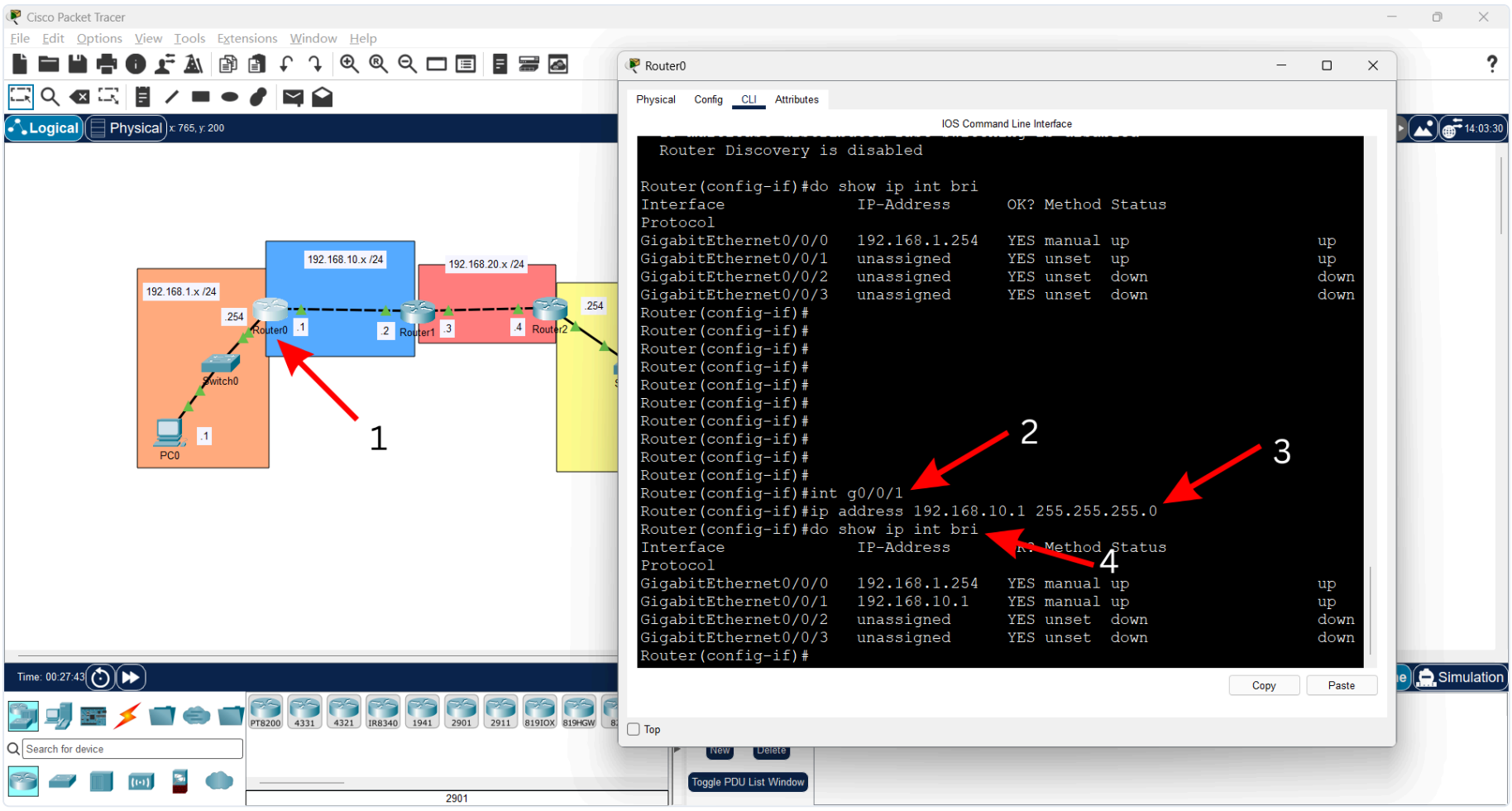
Step 3: Router Interface Provisioning

> **Instructor Facilitation Tip:** Students frequently forget to enter `no shutdown` under physical interfaces. If links remain red, guide students to execute `show ip interface brief` and inspect port Status.

1. Route0 Configuration:

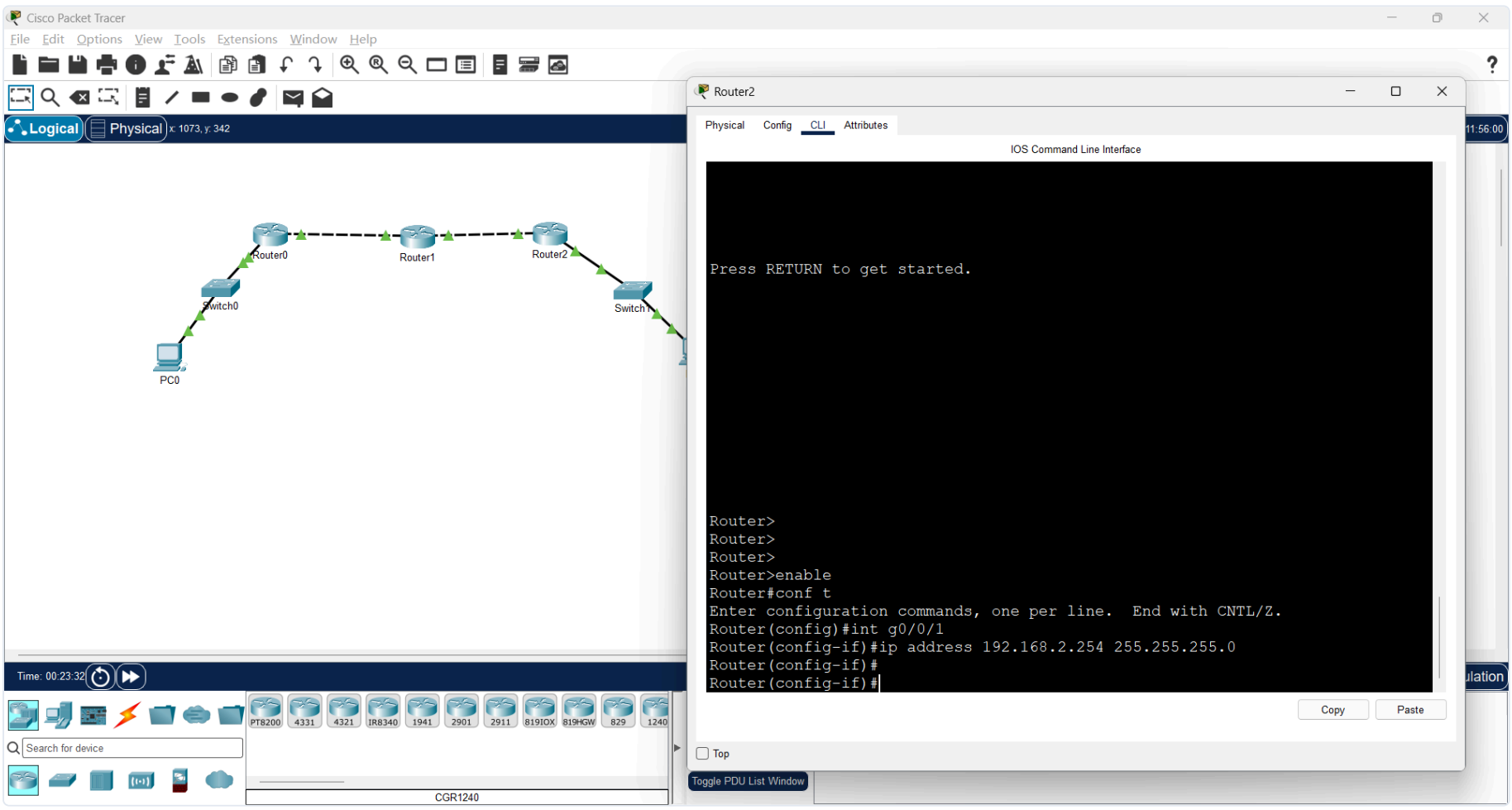
```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.1.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

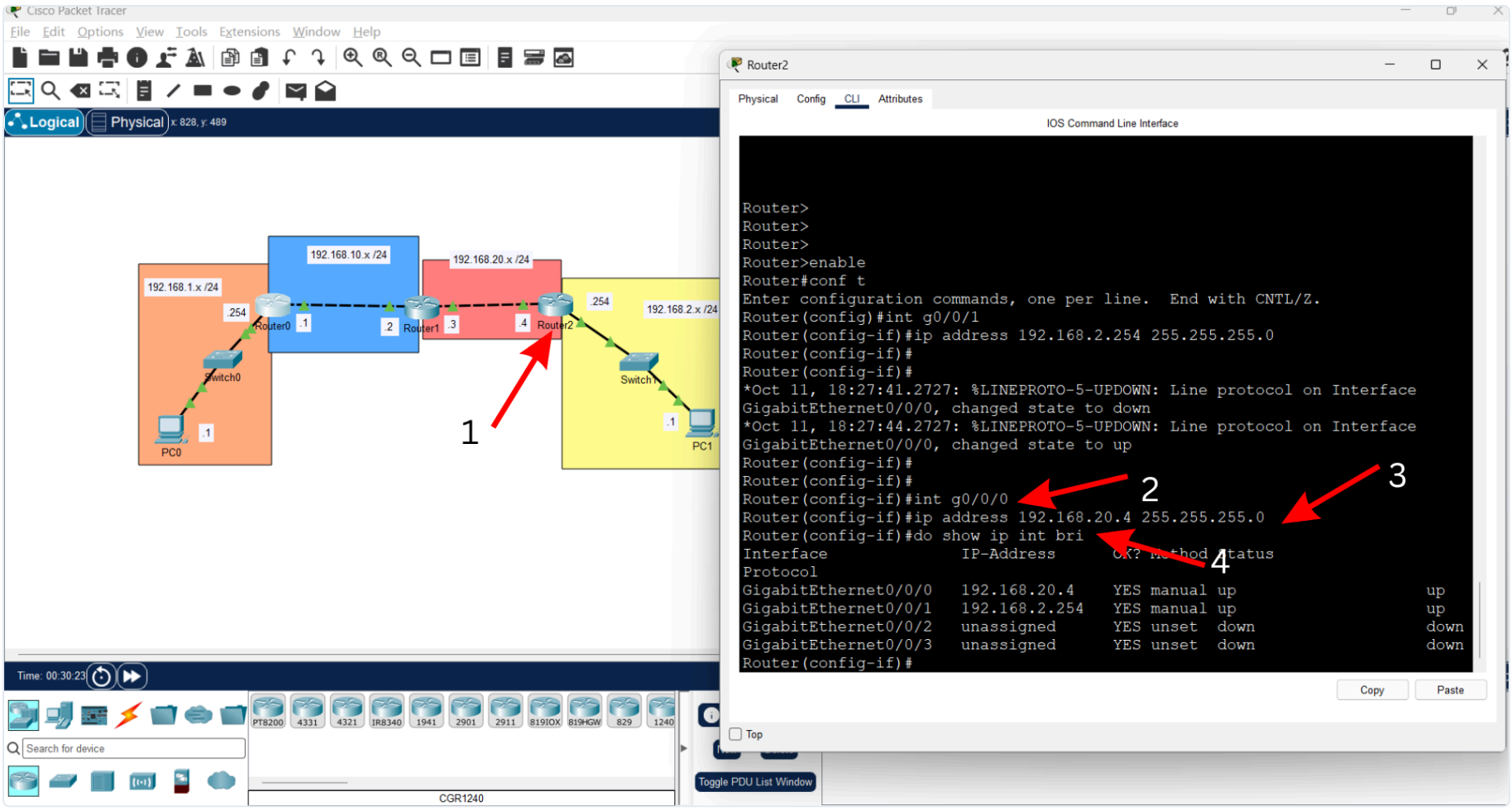




2. Route2 Configuration:

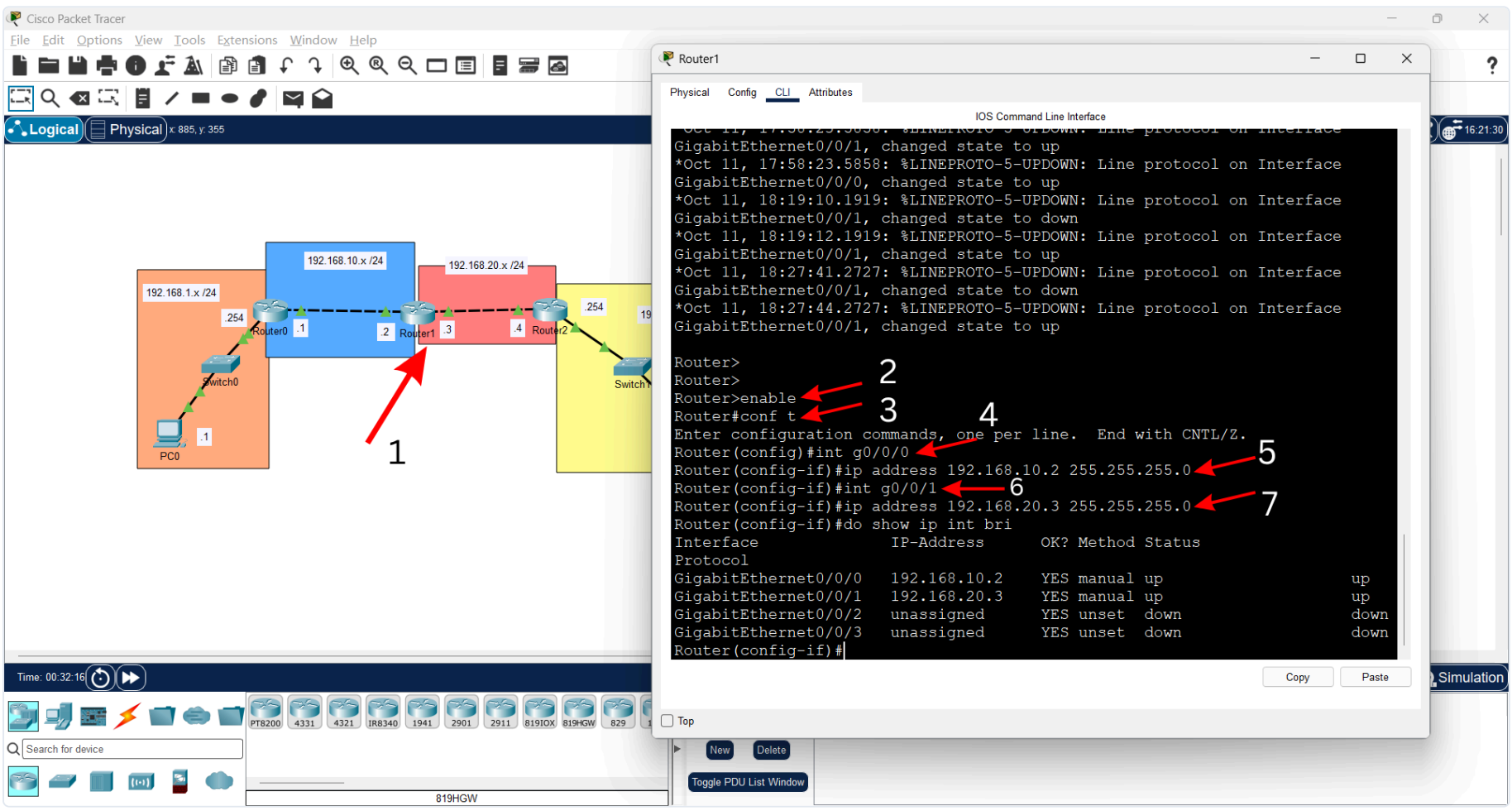
```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.2.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.20.4 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```





3. Route1 Configuration:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.20.3 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

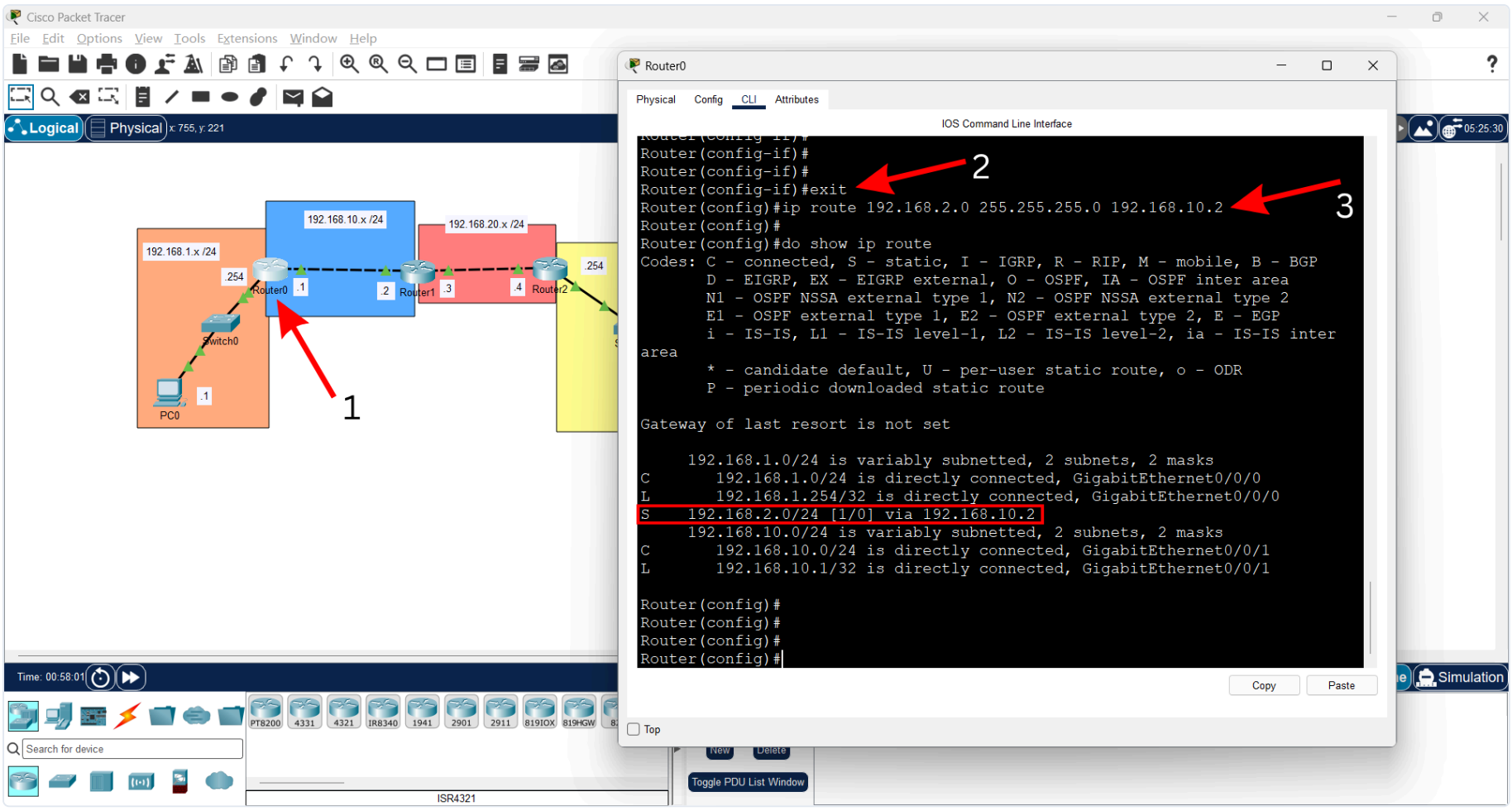


Step 4: Static Route Configuration & Routing Table Inspection

1. On **Route0**, add static route to Network B (192.168.2.0/24):

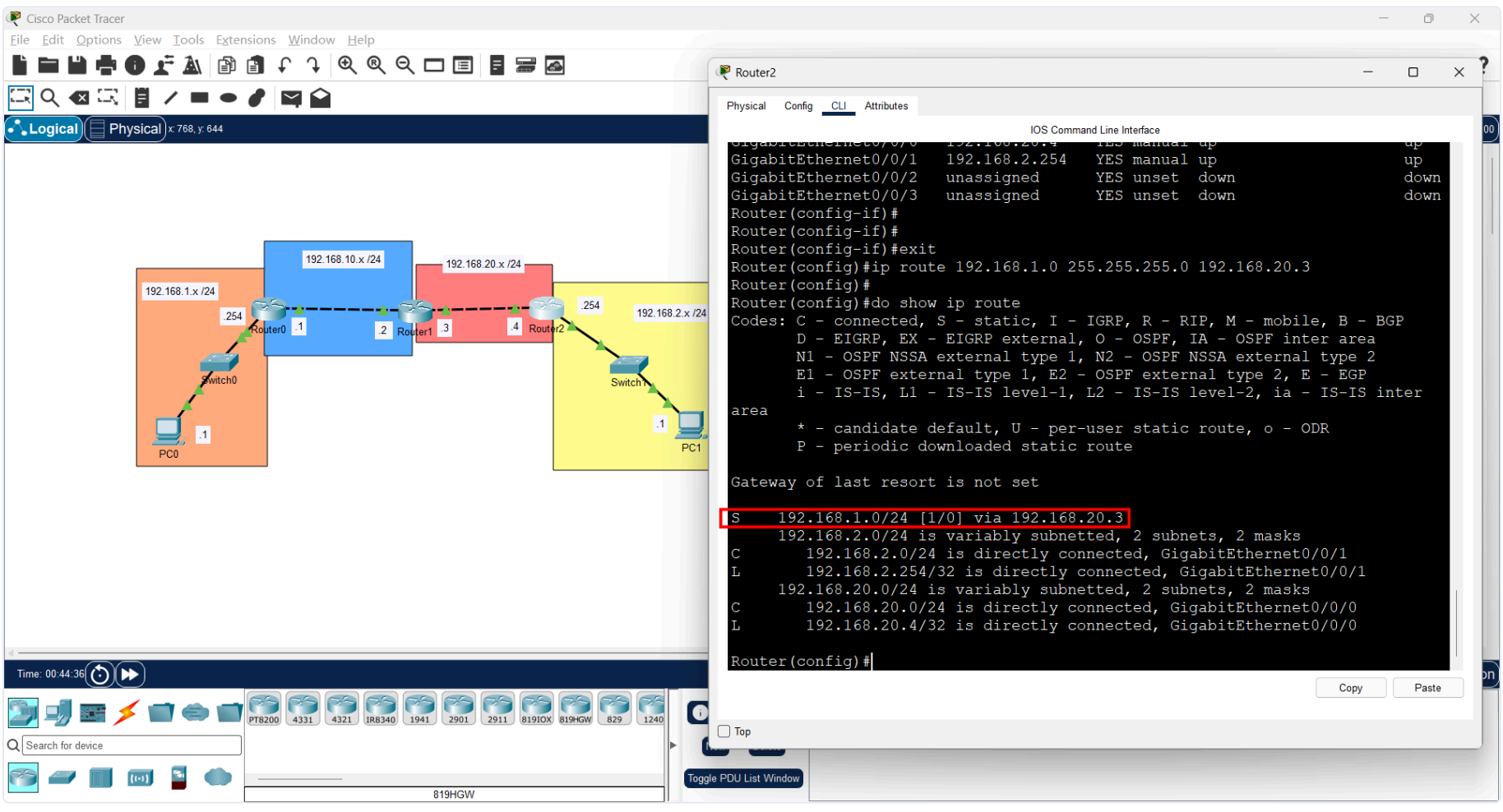
```
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.10.2
Router(config)# do show ip route
```

> **Expected Output on Route0:** Entry `S 192.168.2.0/24 [1/0] via 192.168.10.2` appears in the routing table (RIB).



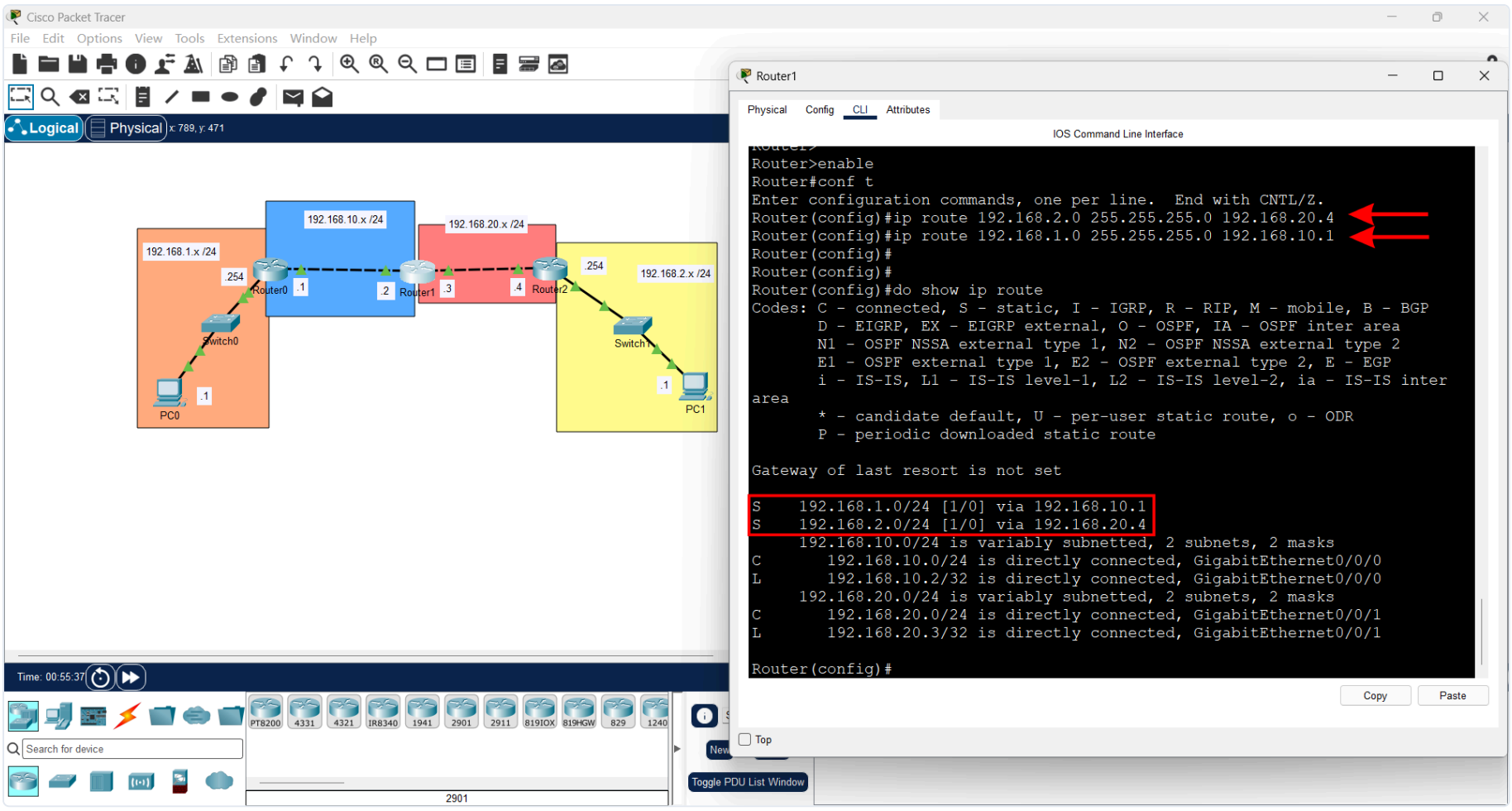
2. On **Route2**, add return static route to Network A (192.168.1.0/24):

```
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.20.3
Router(config)# do show ip route
```



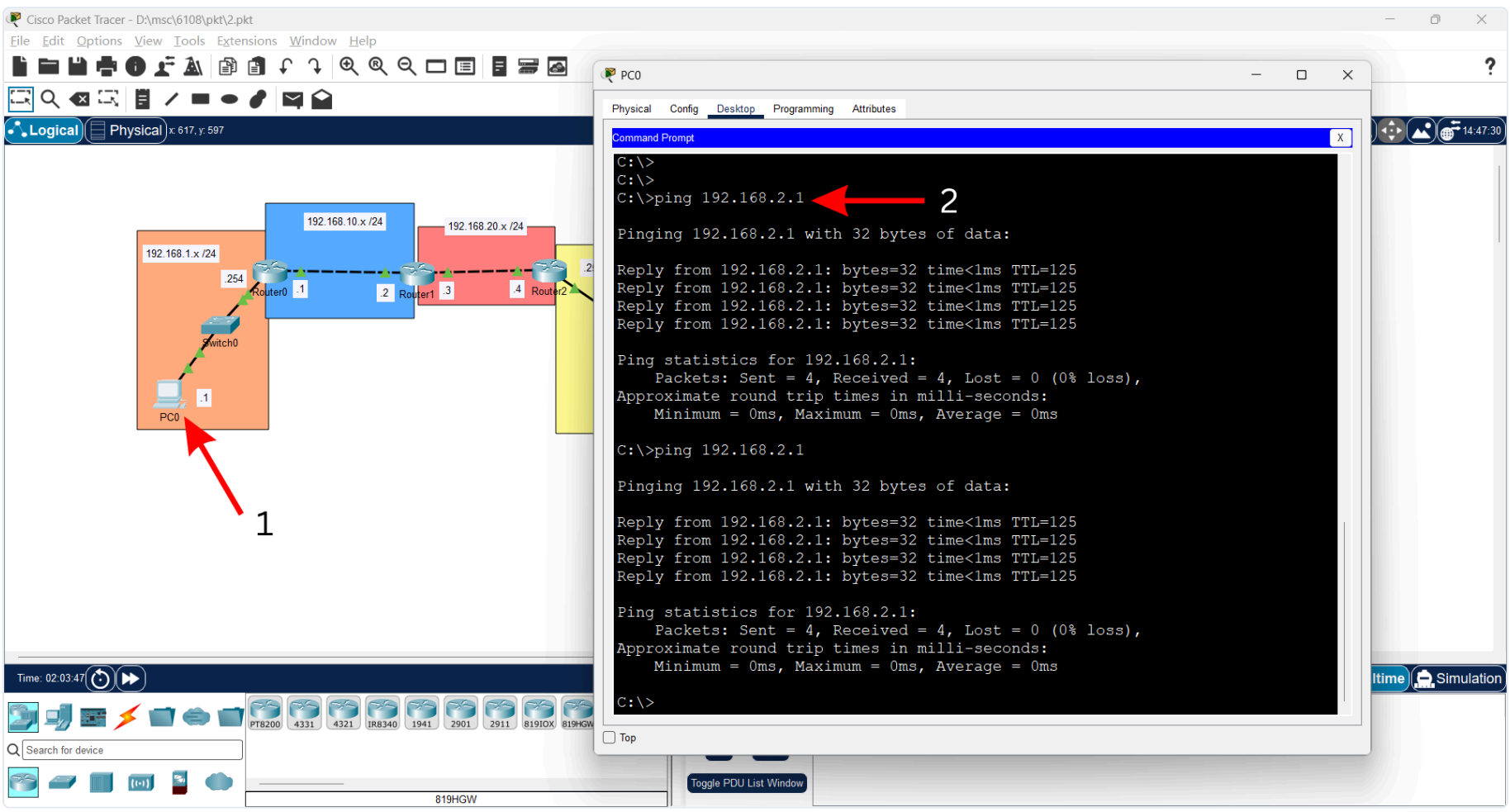
3. On **Route1**, add static routes in both directions:

```
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.20.4
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)# do show ip route
```



Step 5: End-to-End Connectivity Verification

- From PC0 Command Prompt, execute: `ping 192.168.2.1` .
- **Expected Output:** 4 packets sent, 4 received (0% loss).



4. Lab 2 Question Solutions (Grading Guide)

Question 1: Default Gateway Function

Why must PC0 configure a Default Gateway (192.168.1.254) in this lab to reach PC1, whereas in Lab 1 no Default Gateway was required? Explain how host routing behavior changes when the destination IP is on a different subnet.

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Direct vs. Indirect Delivery (2 Marks):* In Lab 1, PC0 and PC1 were on the same subnet (`192.168.1.0/24`). The host directly resolved the target MAC address via ARP and transmitted at Layer 2 without intermediate routing.
> 2. *Gateway Delegation (3 Marks):* In Lab 2, PC1 is on a remote subnet (`192.168.2.0/24`). When the destination IP is outside the local network, host IP logic delegates forwarding to the Default Gateway (`192.168.1.254`). PC0 resolves the MAC address of Route0's interface and encapsulates the packet for Layer 3 transit.

Question 2: Asymmetric Routing & Return Path Omission

If Route0 and Route1 are configured properly, but the return route command on Route2 is omitted, what error will PC0 display when pinging PC1? Why?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Symptom (2 Marks):* PC0 displays `Request timed out` (100% loss).
> 2. *Root Cause (3 Marks):* Network transmission is bidirectional. The outbound ICMP Echo Request successfully reaches PC1. PC1 responds with an ICMP Echo Reply destined for `192.168.1.1` and hands it to Route2. Because Route2 lacks a route back to `192.168.1.0/24`, it drops the reply packet.

Question 3: Missing Intermediate Route

What would happen if the static routes on intermediate router Route1 were completely missing? Would PC0 receive an ICMP Echo Reply from PC1? Explain the forwarding failure point.

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Failure Point (2 Marks):* Communication fails at Route1; no reply is received.
> 2. *Packet Drop Explanation (3 Marks):* Route0 successfully sends the packet to Route1 (`192.168.10.2`). Route1 looks up destination `192.168.2.0/24` in its routing table. Without the static route pointing to Route2 (`192.168.20.4`), Route1 discards the packet immediately as unroutable.

Lab 3: VLAN Segmentation & 802.1Q Trunking

1. Pedagogical Scenario & Teaching Objectives

Participants configure Virtual Local Area Networks (VLANs) across two interconnected Layer 2 switches. Students explore how VLANs isolate broadcast domains within shared physical hardware and configure an 802.1Q Trunk link to multiplex multiple VLANs across switches.

- Create VLAN 10 and VLAN 20 on Cisco Catalyst 2960 switches.
- Configure Access switch ports to isolate end hosts into distinct broadcast domains.
- Configure the inter-switch link as an 802.1Q Trunk allowing VLANs 10 and 20.
- Verify that same-VLAN hosts communicate across switches while cross-VLAN traffic is blocked at Layer 2.

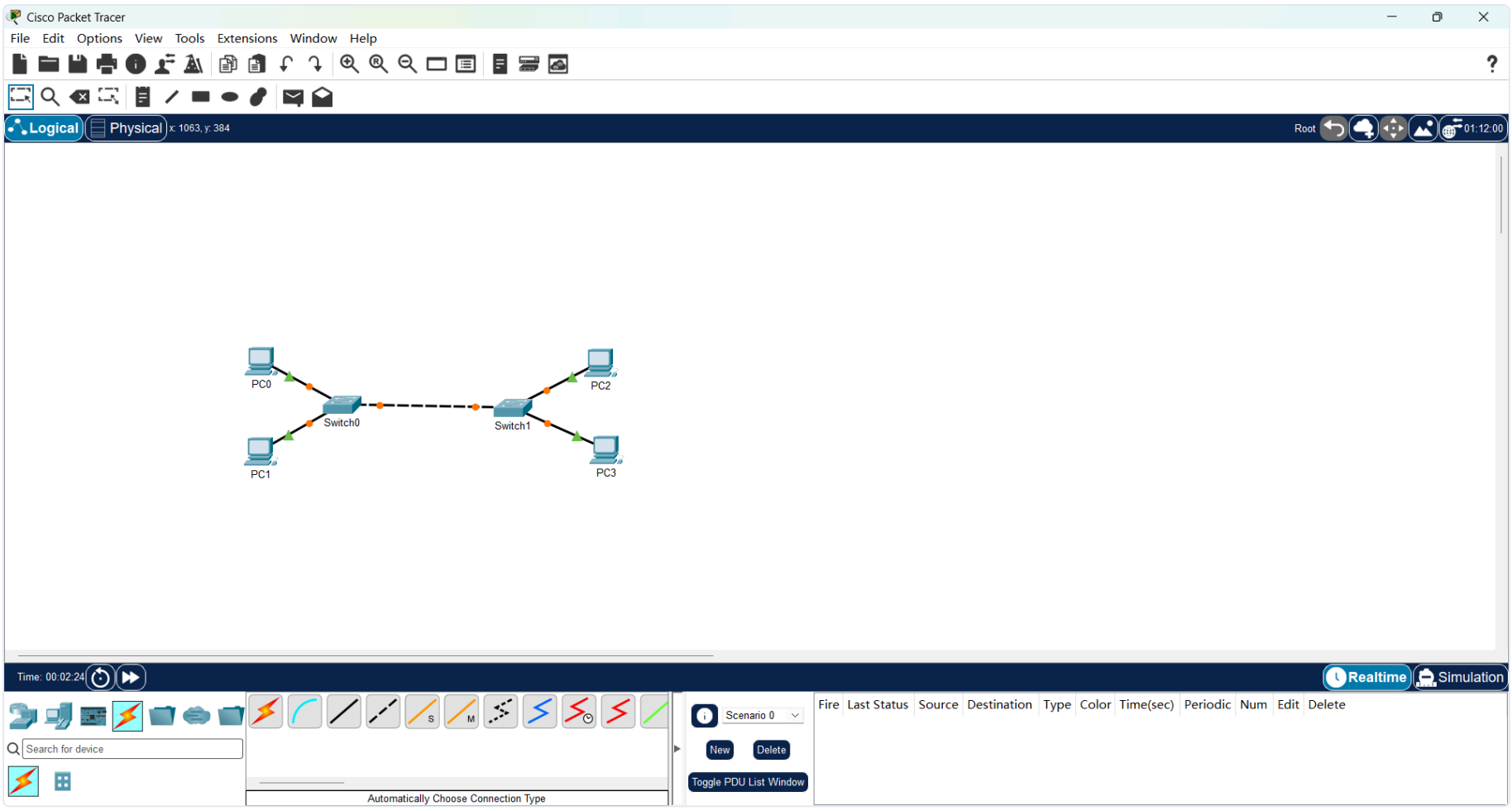
2. Master Addressing & Port Assignment Matrix

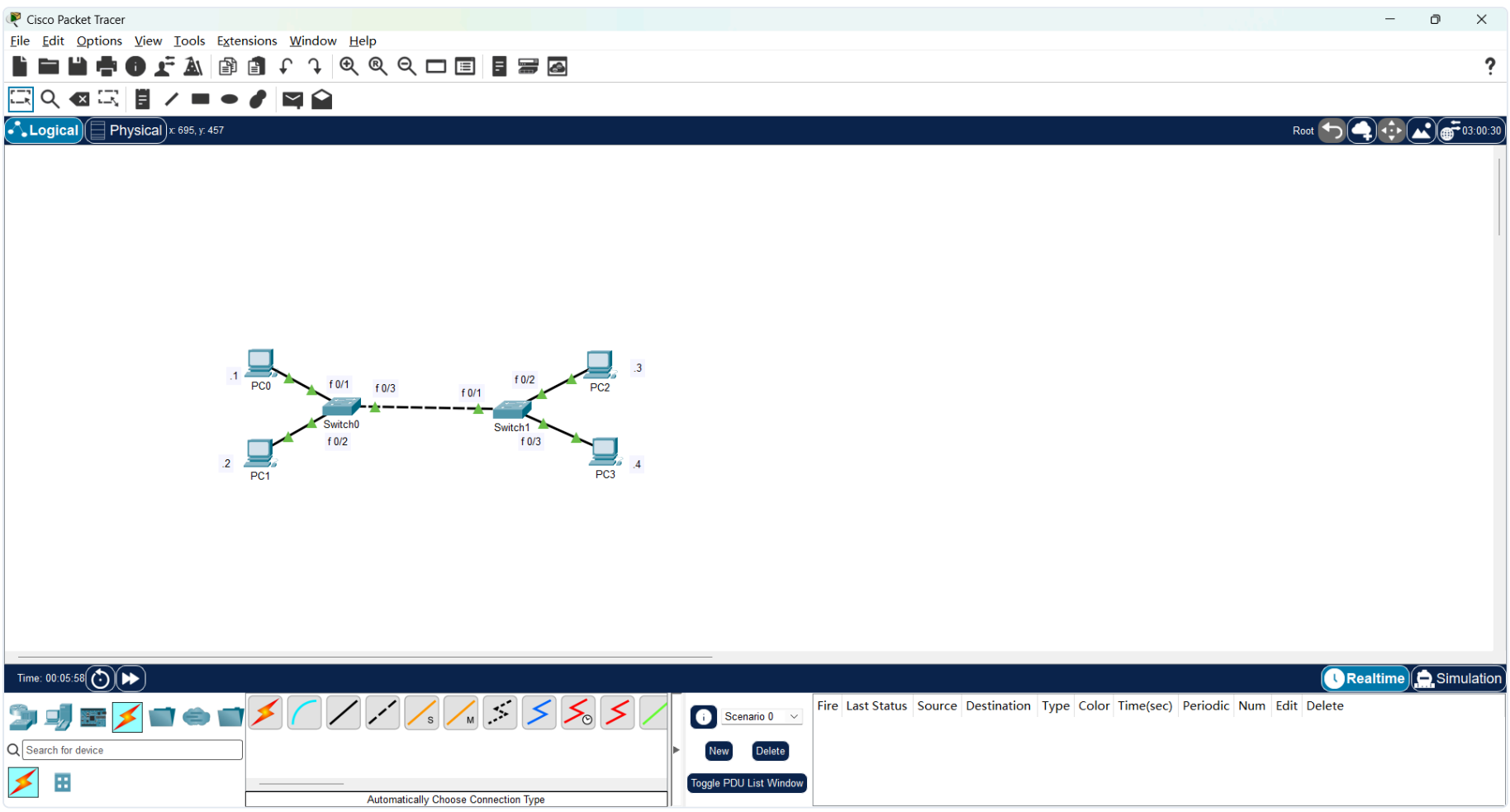
Device	Interface	IP Address	Subnet Mask	VLAN Assignment	Connected Switch Port
PC0	FastEthernet0	192.168.1.1	255.255.255.0	VLAN 10	Switch0 Fa0/1
PC1	FastEthernet0	192.168.1.2	255.255.255.0	VLAN 20	Switch0 Fa0/2
PC2	FastEthernet0	192.168.1.3	255.255.255.0	VLAN 10	Switch1 Fa0/2
PC3	FastEthernet0	192.168.1.4	255.255.255.0	VLAN 20	Switch1 Fa0/3
Trunk Link	Fa0/3 ↔ Fa0/1	N/A	N/A	VLANs 10, 20 Allowed	Switch0 Fa0/3 ↔ Switch1 Fa0/1

3. Step-by-Step Facilitation & Solution Checkpoints

Step 1: Placement, Auto-Cabling & Host IP Setup

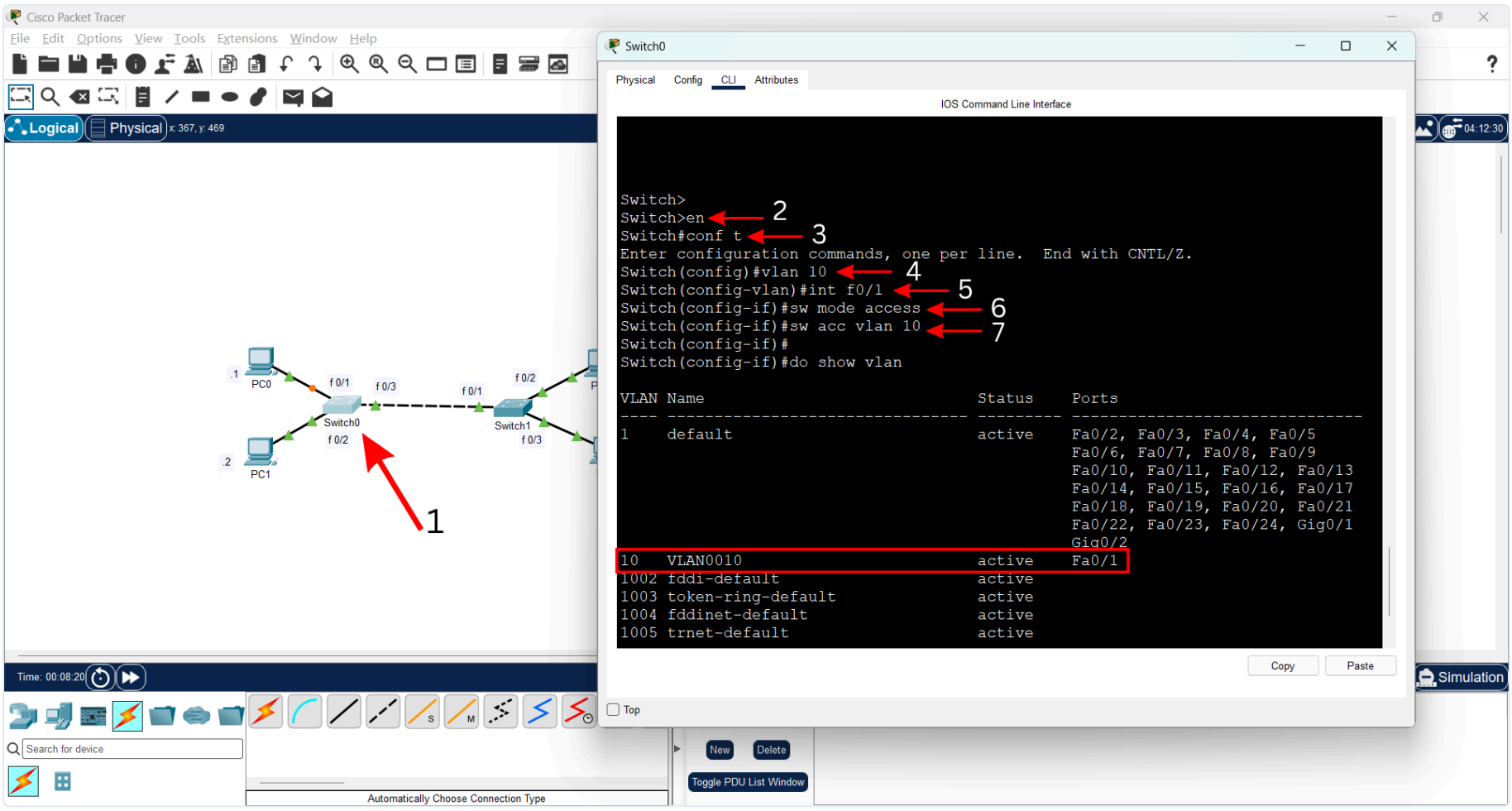
- Deploy 4 PCs and 2 switches (Catalyst 2960). Connect devices using auto cabling.
- On each PC, set static IPs via **Config** → **FastEthernet0** as per table.





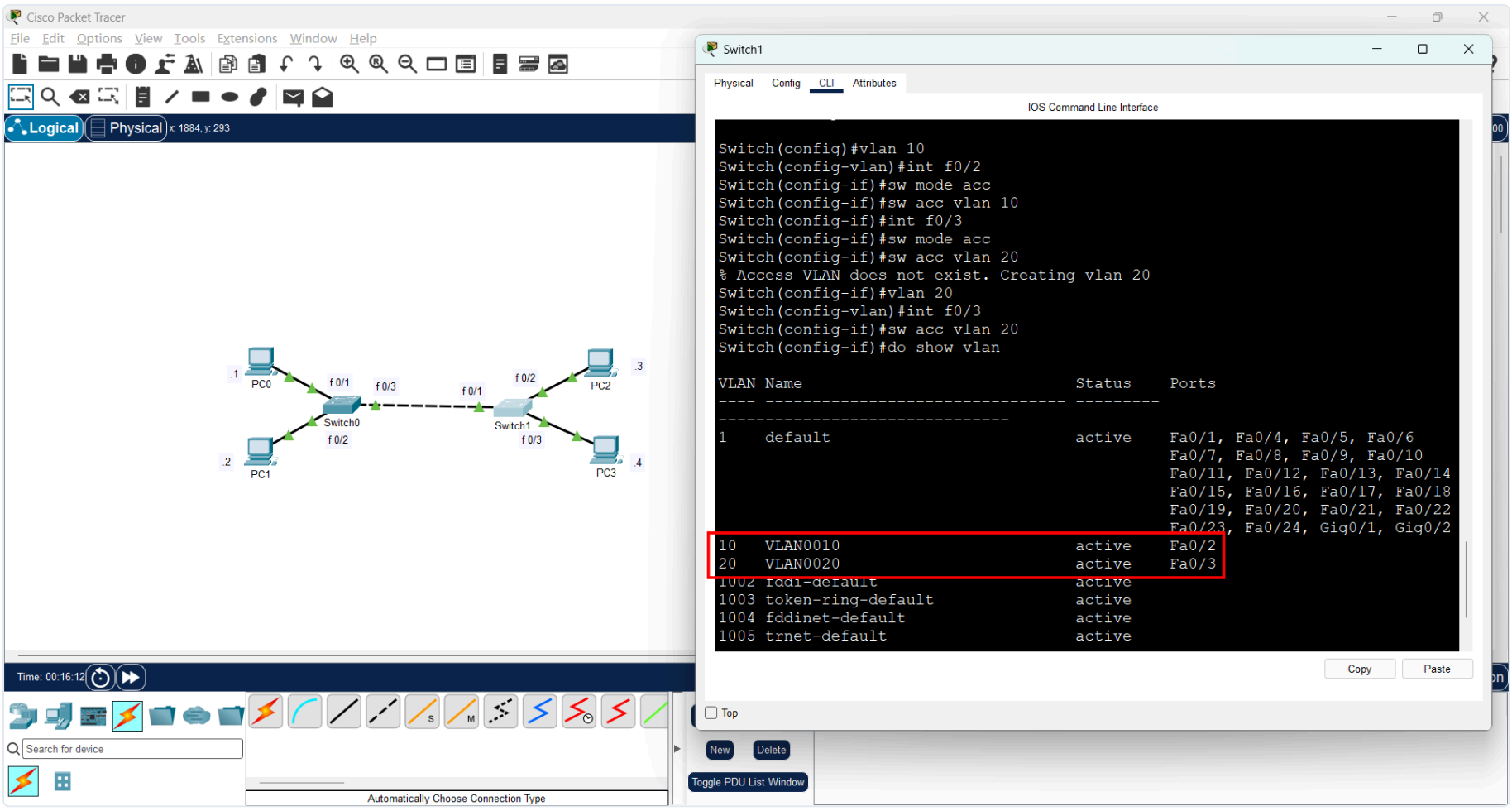
Step 2: Configure Access VLANs on Switch0

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# exit
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# vlan 20
Switch(config-vlan)# exit
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# do show vlan
```



Step 3: Configure Access VLANs on Switch1

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config)# vlan 20
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# interface fastethernet 0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# do show vlan
```

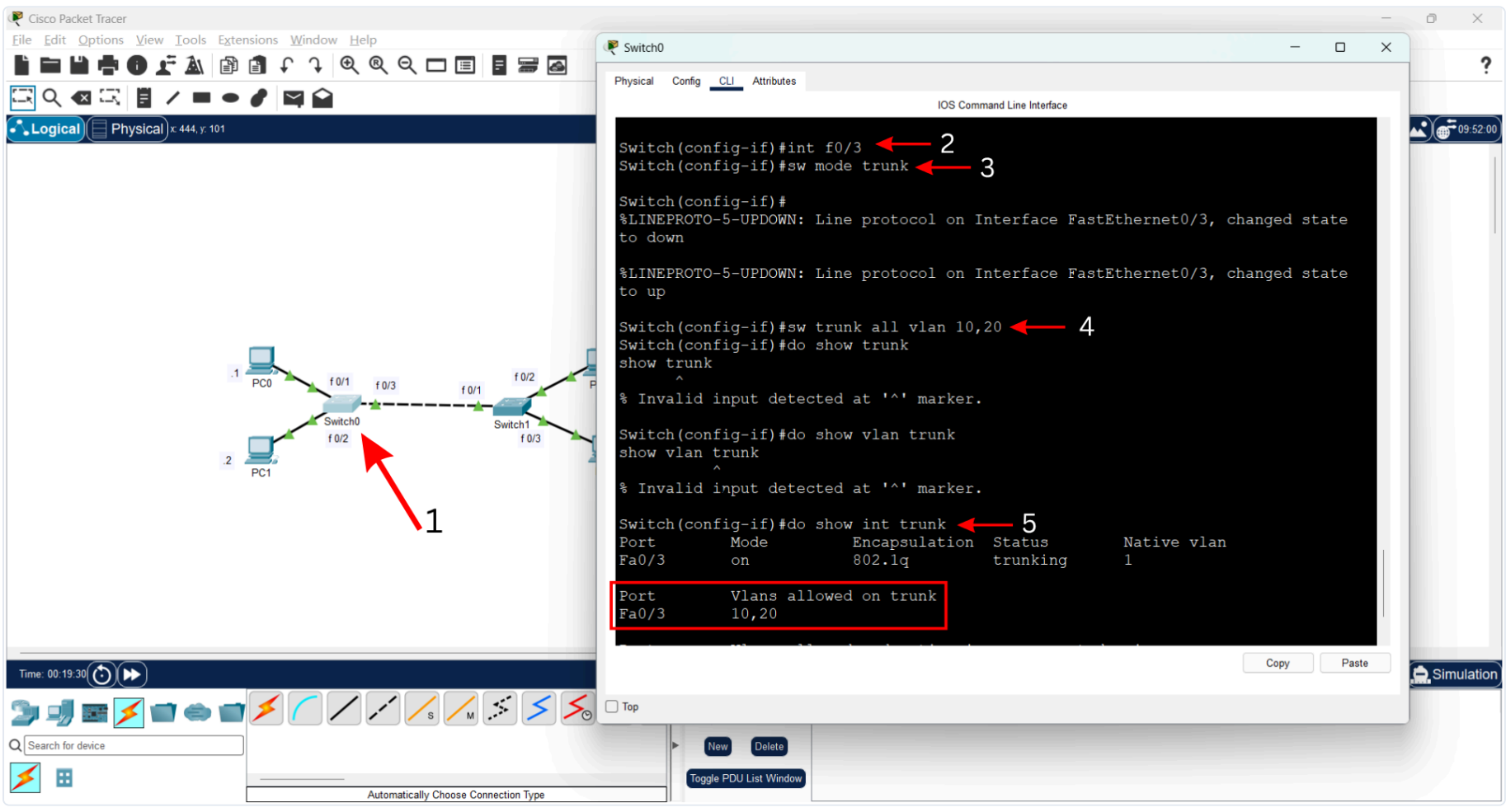


Step 4: Configure Trunk Mode on Inter-Switch Links

> **Instructor Facilitation Tip:** Ensure students run `show interfaces trunk` to verify that Fa0/3 (Switch0) and Fa0/1 (Switch1) show status "trunking" and list VLANs 10, 20 under allowed VLANs.

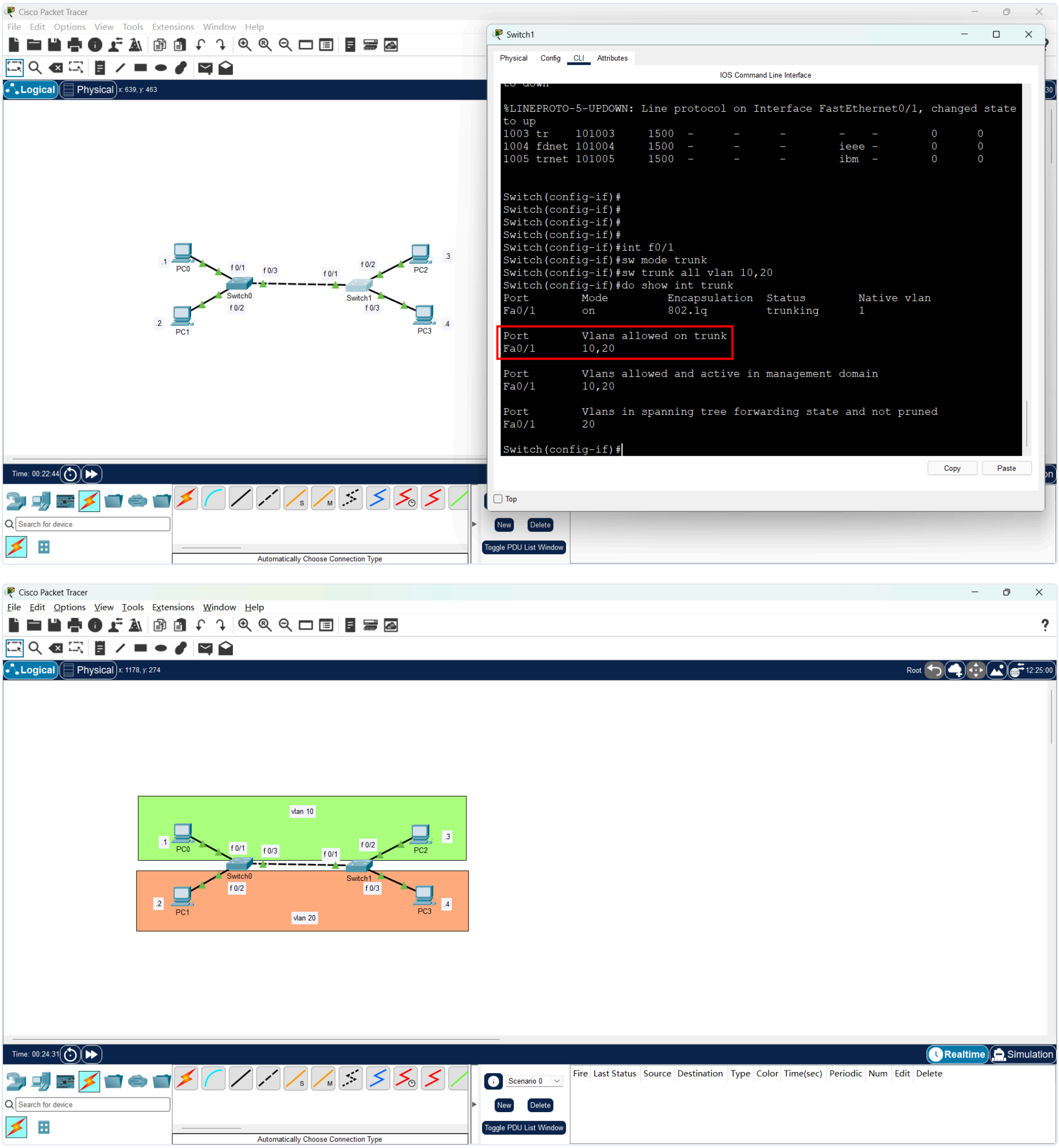
1. Switch0 Trunk Configuration:

```
Switch(config)# interface fastethernet 0/3
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# do show interfaces trunk
```



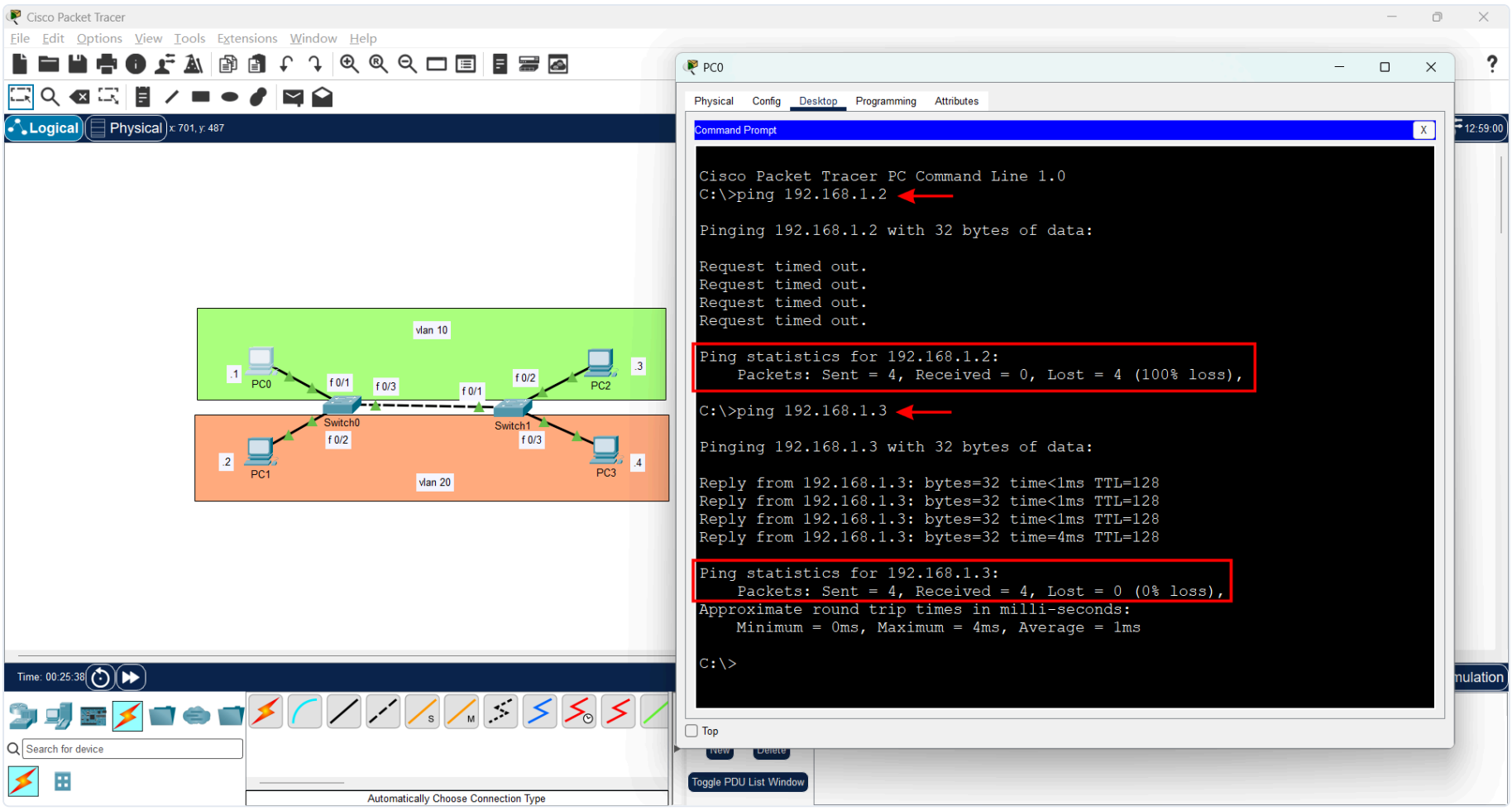
2. Switch1 Trunk Configuration:

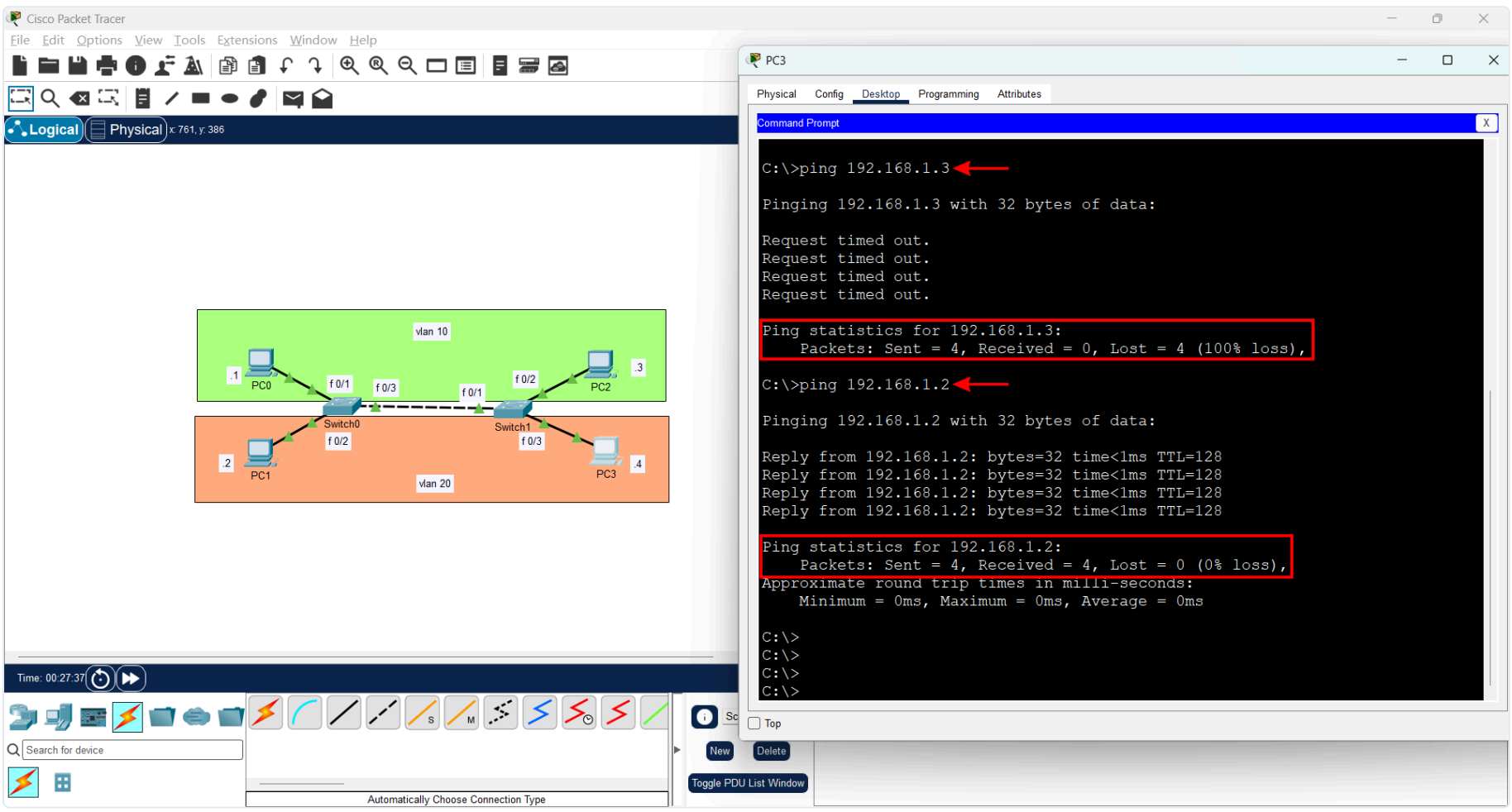
```
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# do show interfaces trunk
```



Step 5: Testing Isolation and Connectivity

- PC0 (VLAN 10): Ping PC2 (VLAN 10) → Success (0% loss); Ping PC1 (VLAN 20) → Request timed out (Blocked).
- PC3 (VLAN 20): Ping PC1 (VLAN 20) → Success (0% loss); Ping PC2 (VLAN 10) → Request timed out (Blocked).





4. Lab 3 Question Solutions (Grading Guide)

Question 1: VLAN Broadcast Isolation

Although PC0 (192.168.1.1) and PC1 (192.168.1.2) are in the same IP subnet (192.168.1.0/24) and connected to the same switch, why do they fail to communicate? Explain the role of VLAN tags.

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Broadcast Domain Partitioning (2 Marks):* VLANs divide a physical switch into distinct logical broadcast domains. Switch0 isolates port Fa0/1 into VLAN 10 and Fa0/2 into VLAN 20.
> 2. *ARP Blocking Mechanism (3 Marks):* When PC0 attempts to ping PC1, it must resolve PC1's MAC address via an ARP broadcast. The switch tags this broadcast with VLAN 10 and strictly forwards it only to ports belonging to VLAN 10. PC1 (on VLAN 20) never receives the ARP request, so no IP packet can be encapsulated.

Question 2: Trunk Port Function

What is the primary difference between a switch port configured in Access mode versus Trunk mode? Why is a Trunk port necessary between Switch0 and Switch1?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Access vs. Trunk Distinction (2 Marks):* An Access port belongs to a single VLAN and transmits untagged Ethernet frames to end hosts. A Trunk port carries traffic for multiple VLANs across a single physical link by inserting an IEEE 802.1Q tag.
> 2. *Multiplexing Necessity (3 Marks):* Without a Trunk link, connecting VLAN 10 and VLAN 20 across switches would require two separate physical cables (one for each VLAN). The Trunk multiplexes VLAN 10 and VLAN 20 traffic over a single cable, conserving switch ports.

Lab 4: Multi-Subnet Static Routing & Standard Access Control Lists (ACL)

1. Pedagogical Scenario & Teaching Objectives

Participants construct a routed network spanning four subnets across two routers and implement security filtering using Standard Access Control Lists (ACLs). Students verify baseline reachability before applying ACLs to enforce host-level and subnet-level security policies.

- Establish static routing across four distinct network segments via two routers.
- Verify baseline reachability across all endpoints prior to deploying security rules.
- Configure Standard Named ACLs matching individual source hosts and entire subnets (wildcard mask).
- Apply ACL filters to router outbound interfaces (`out`) and verify traffic permit/deny enforcement.

2. Master Architecture & Addressing Matrix

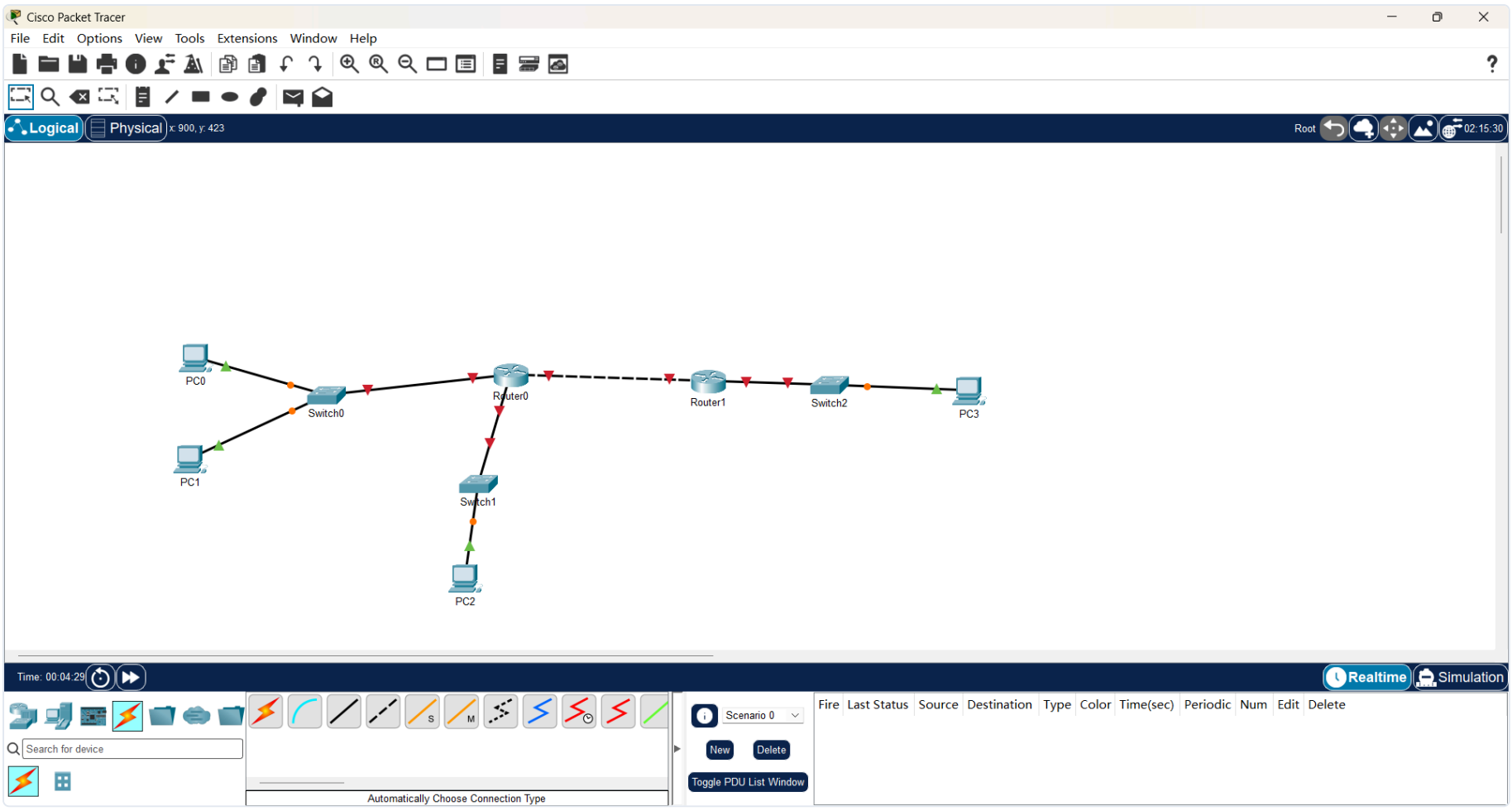
> **Four Subnet Partitions (/24 Prefix):**
> - **Orange Zone:** `192.168.1.0/24` | PC0 (.1), PC1 (.2), Router0 G0/0 (.254)
> - **Green Zone:** `192.168.2.0/24` | PC2 (.1), Router0 G0/1 (.254)
> - **Yellow Backbone:** `192.168.10.0/24` | Router0 G0/2 (.1) ↔ Router1 G0/0 (.2)
> - **Purple Zone:** `192.168.3.0/24` | PC3 (.1), Router1 G0/1 (.254)

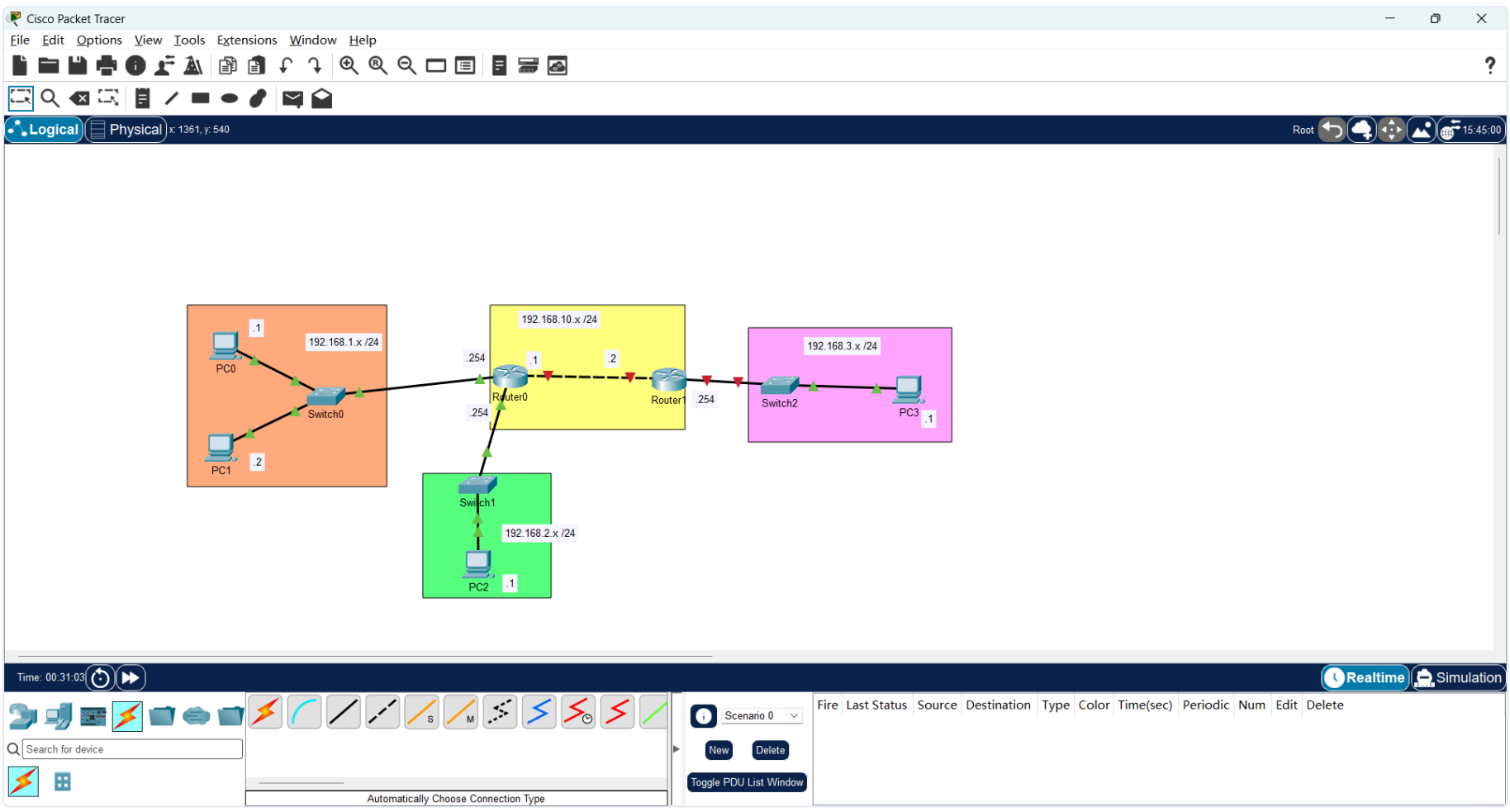
Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.1	255.255.255.0	**192.168.1.254**
PC1	FastEthernet0	192.168.1.2	255.255.255.0	**192.168.1.254**
PC2	FastEthernet0	192.168.2.1	255.255.255.0	**192.168.2.254**
PC3	FastEthernet0	192.168.3.1	255.255.255.0	**192.168.3.254**
Router0	Gig0/0	192.168.1.254	255.255.255.0	LAN Gateway for PC0/PC1
	Gig0/1	192.168.2.254	255.255.255.0	LAN Gateway for PC2
	Gig0/2	192.168.10.1	255.255.255.0	WAN Link toward Router1
Router1	Gig0/0	192.168.10.2	255.255.255.0	WAN Link toward Router0
	Gig0/1	192.168.3.254	255.255.255.0	LAN Gateway for PC3

3. Step-by-Step IOS Commands & Verification Key

Step 1: Construct Topology and Host Configuration

- Deploy 4 PCs, 3 switches, and 2 routers. Connect using **Automatically Choose Connection Type**.
- Configure host IP addresses and Default Gateways via the **Config** tab.

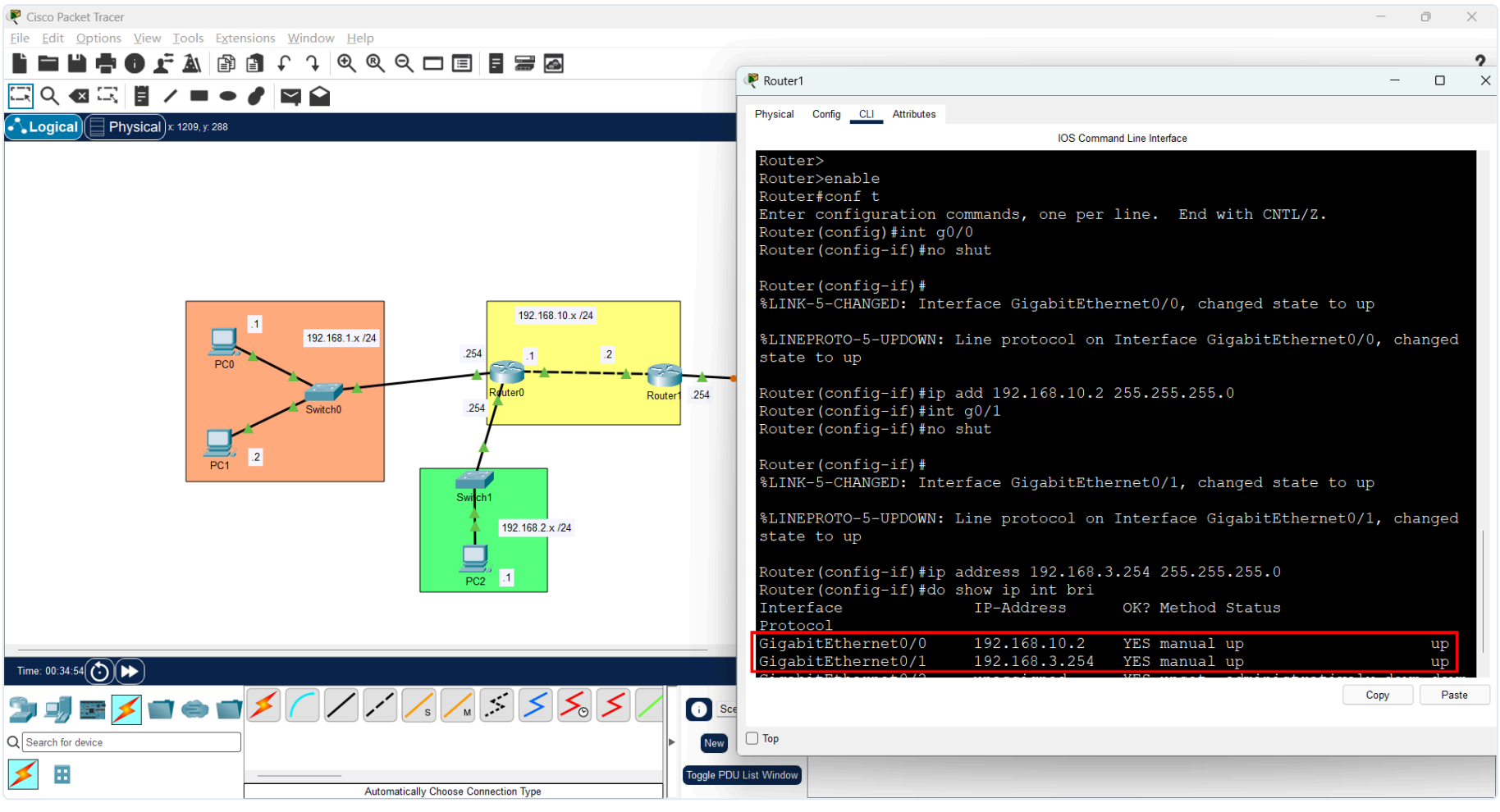




Step 2: Router Interface & Static Route Configuration

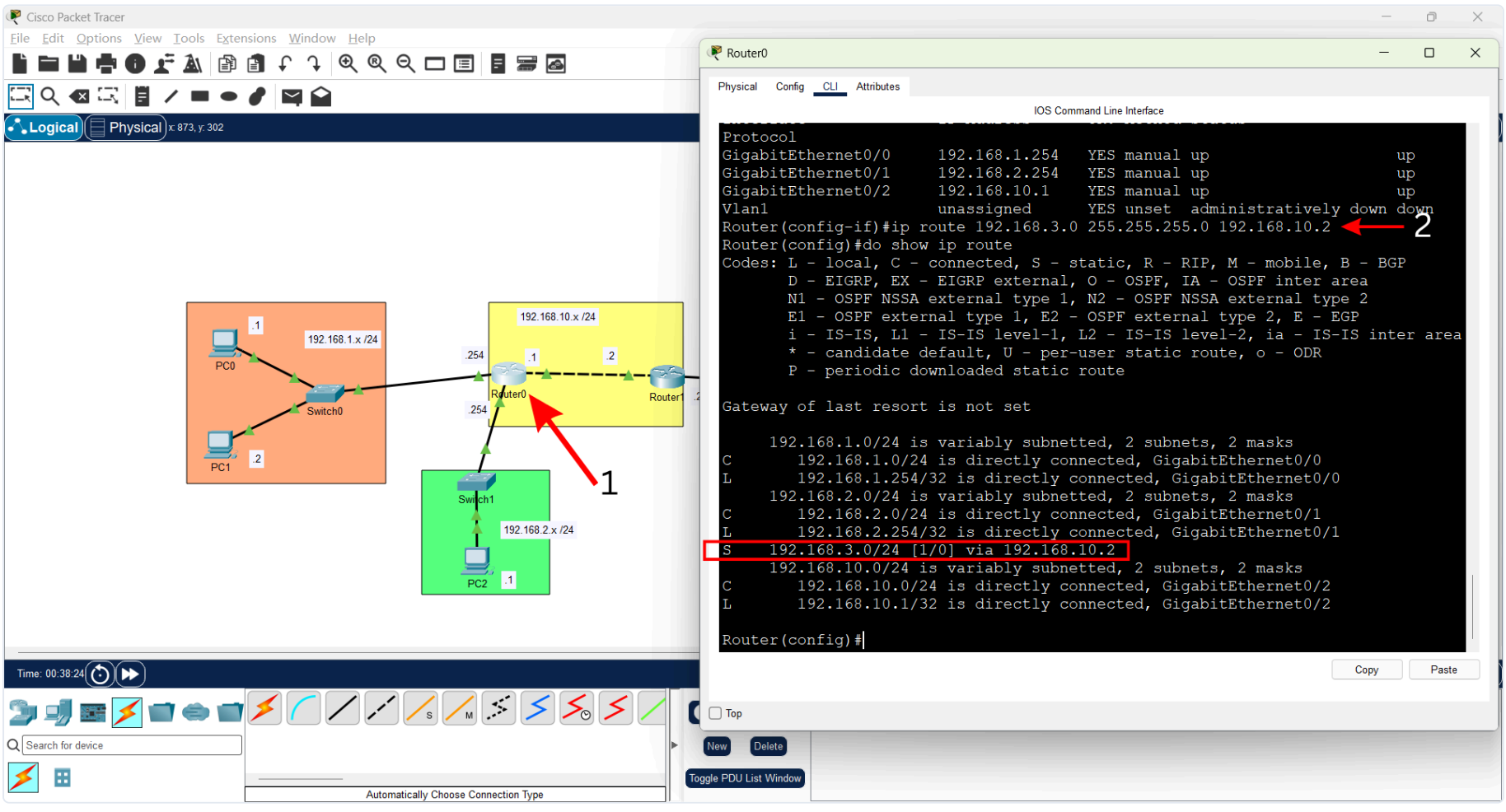
1. Router1 Interface Provisioning:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitethernet 0/1
Router(config-if)# ip address 192.168.3.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# do show ip int bri
```



2. Router0 Interfaces & Static Route to 192.168.3.0/24:

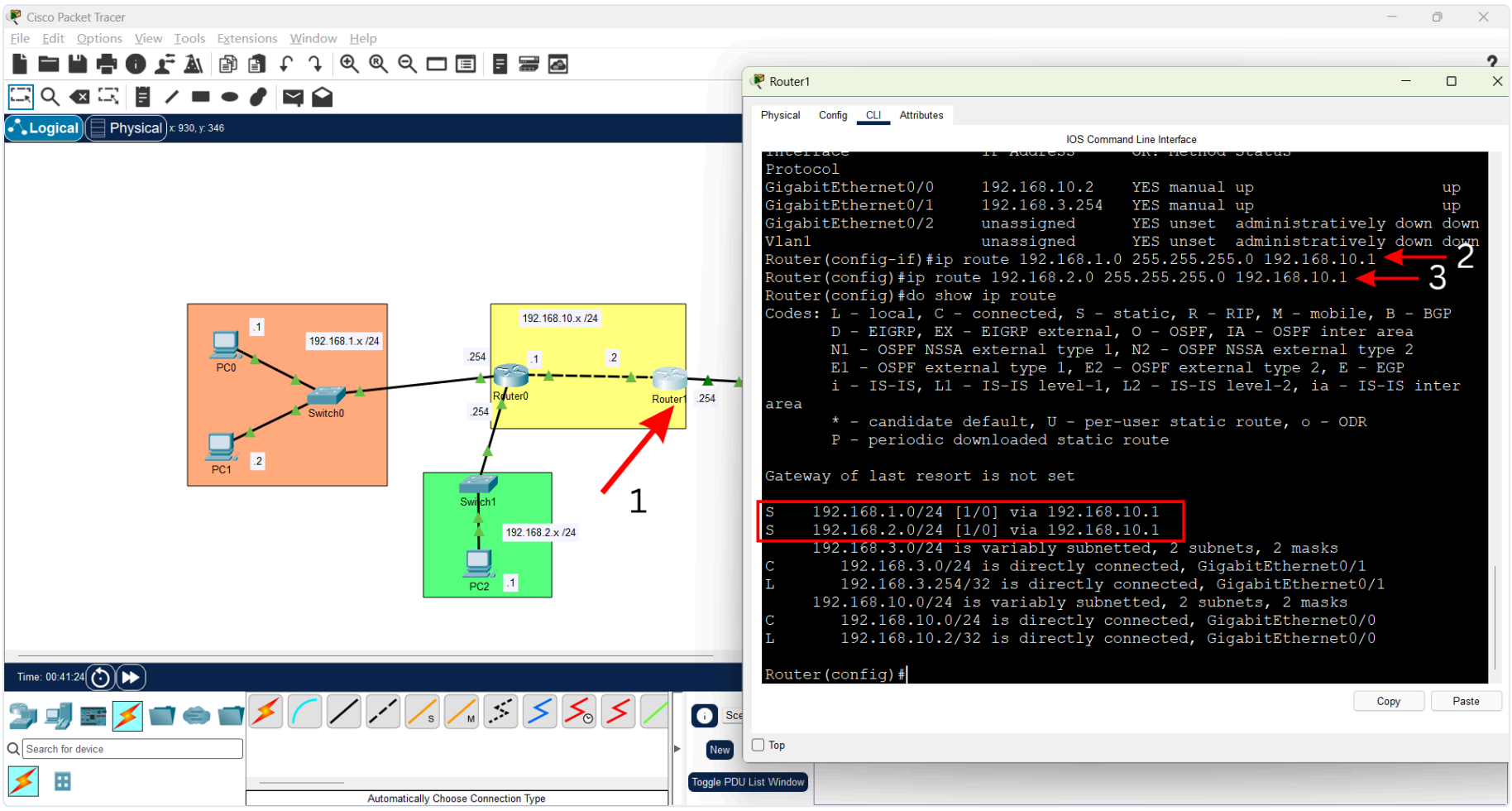
```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0
Router(config-if)# ip address 192.168.1.254 255.255.255.0
Router(config-if)# no shutdown
Router(config)# interface gigabitethernet 0/1
Router(config-if)# ip address 192.168.2.254 255.255.255.0
Router(config-if)# no shutdown
Router(config)# interface gigabitethernet 0/2
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ip route 192.168.3.0 255.255.255.0 192.168.10.2
Router(config)# do show ip route
```



3. Router1 Return Static Routes to 192.168.1.0/24 and 192.168.2.0/24:

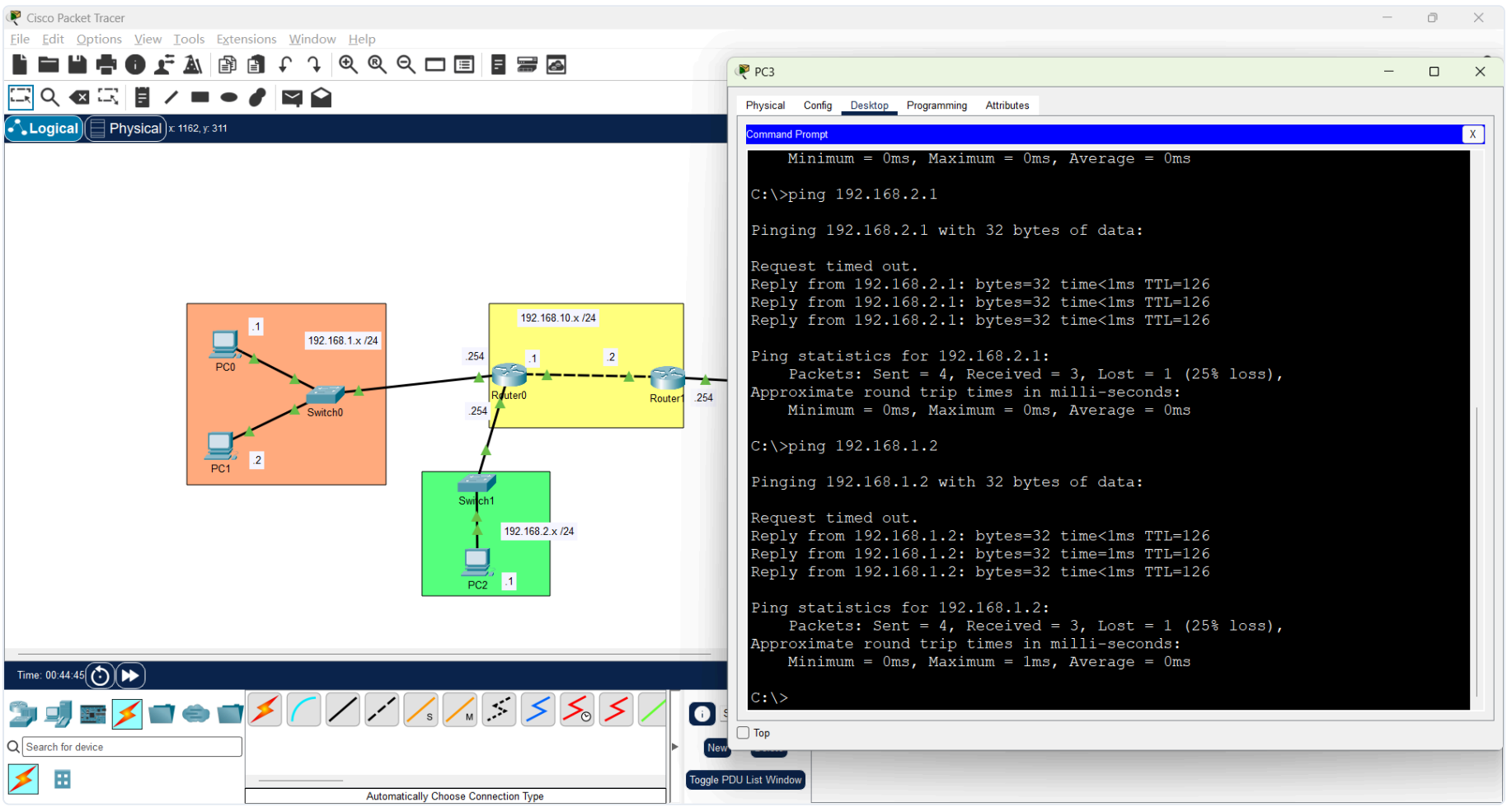
```
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.10.1
```

```
Router(config)# do show ip route
```



Step 3: Baseline Reachability Verification

- From PC3, ping PC2 (`192.168.2.1`) → Success (0% loss).
- From PC3, ping PC1 (`192.168.1.2`) → Success (0% loss).



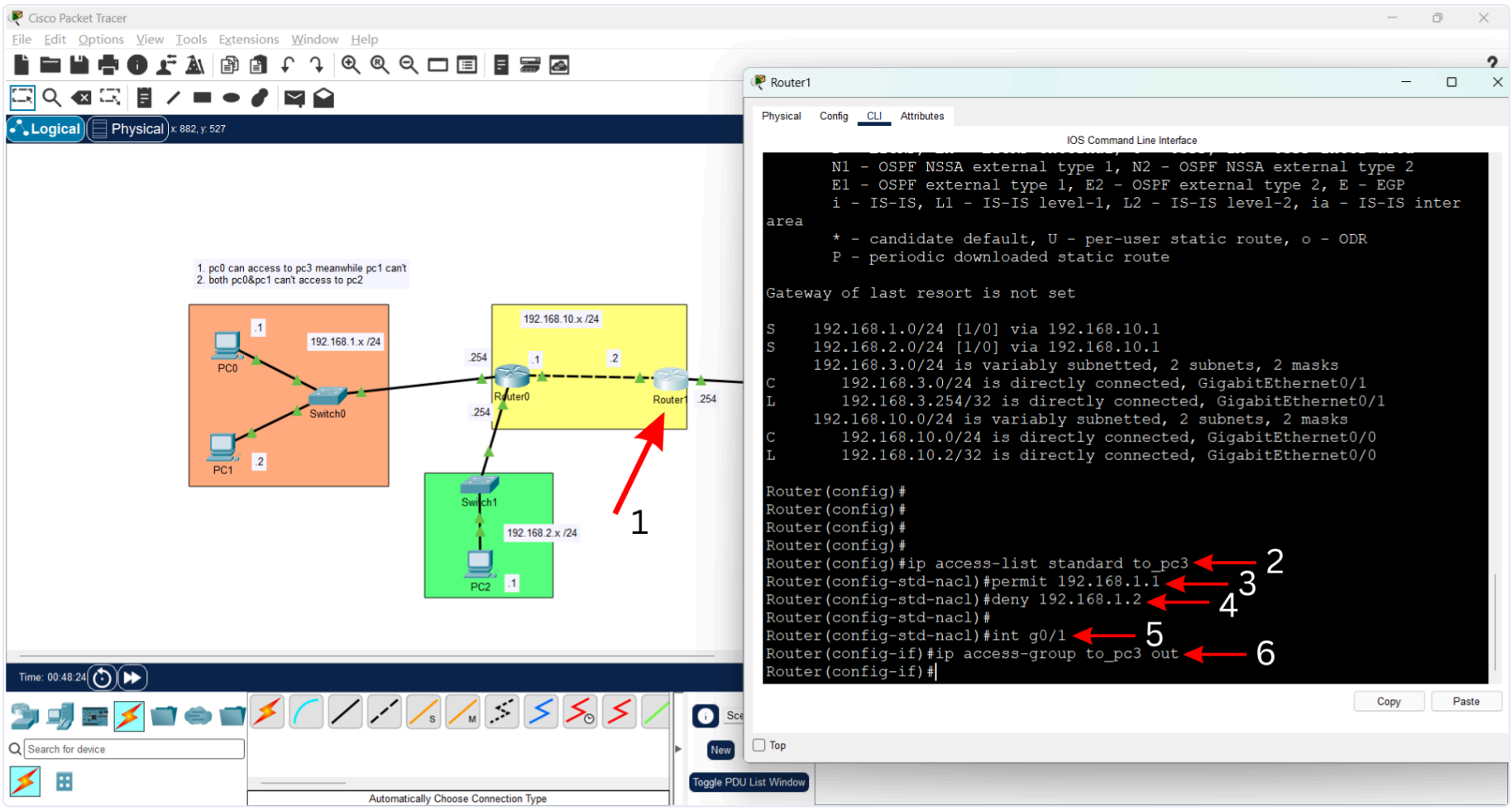
Step 4: Standard ACL Configuration & Interface Binding

> **Instructor Facilitation Tip:** Explain that standard ACLs can only filter by *Source IP*. Emphasize why the ACLs are bound in the **out** direction on the LAN gateway interface closest to the destination.

1. Policy 1 (On Router1, protect PC3):

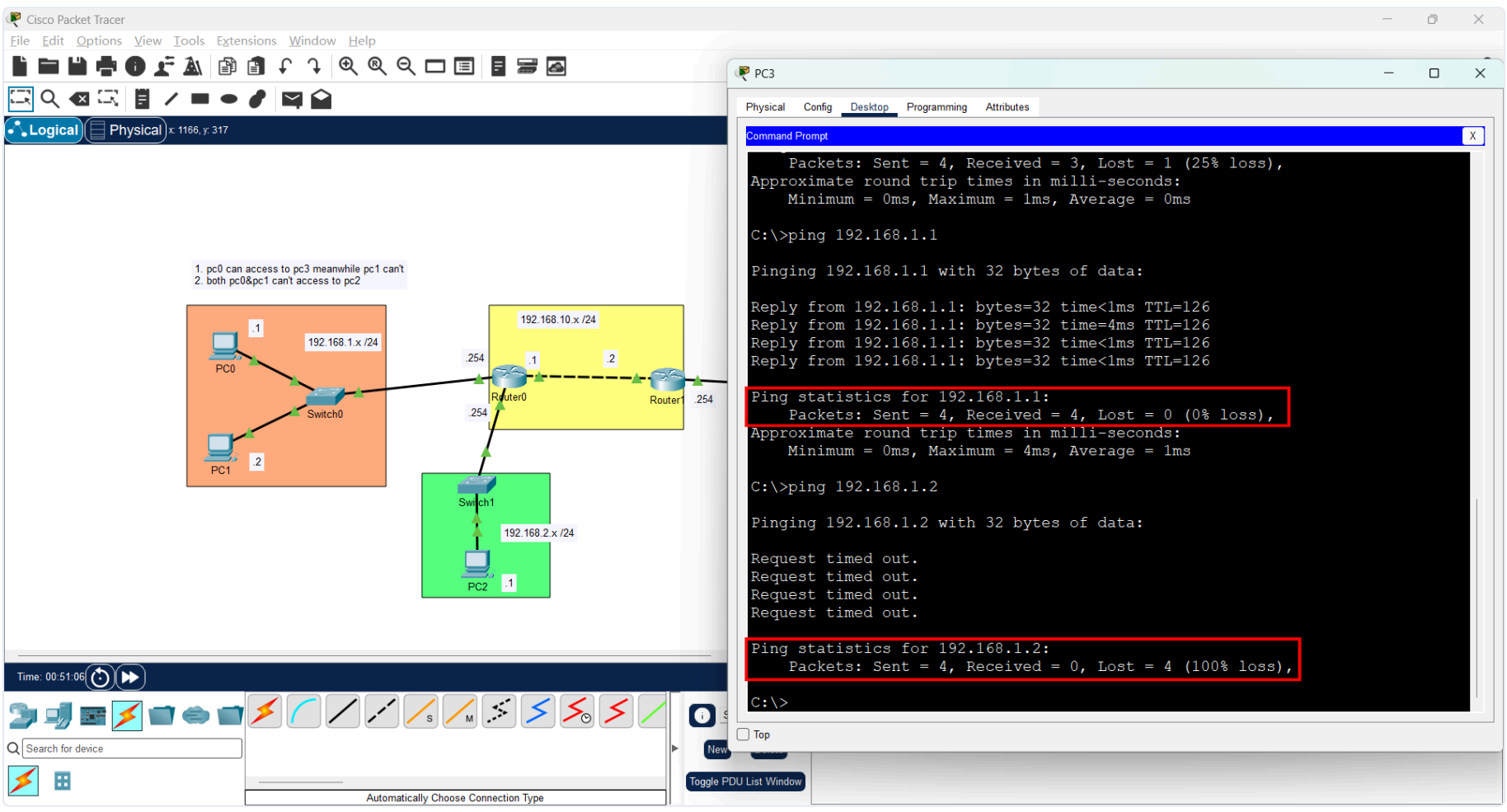
```
Router1(config)# ip access-list standard to_pc3
Router1(config-std-nacl)# permit 192.168.1.1
Router1(config-std-nacl)# deny 192.168.1.2
Router1(config-std-nacl)# exit
Router1(config)# interface gigabitethernet 0/1
```

```
Router1(config-if)# ip access-group to_pc3 out
Router1(config-if)# exit
```



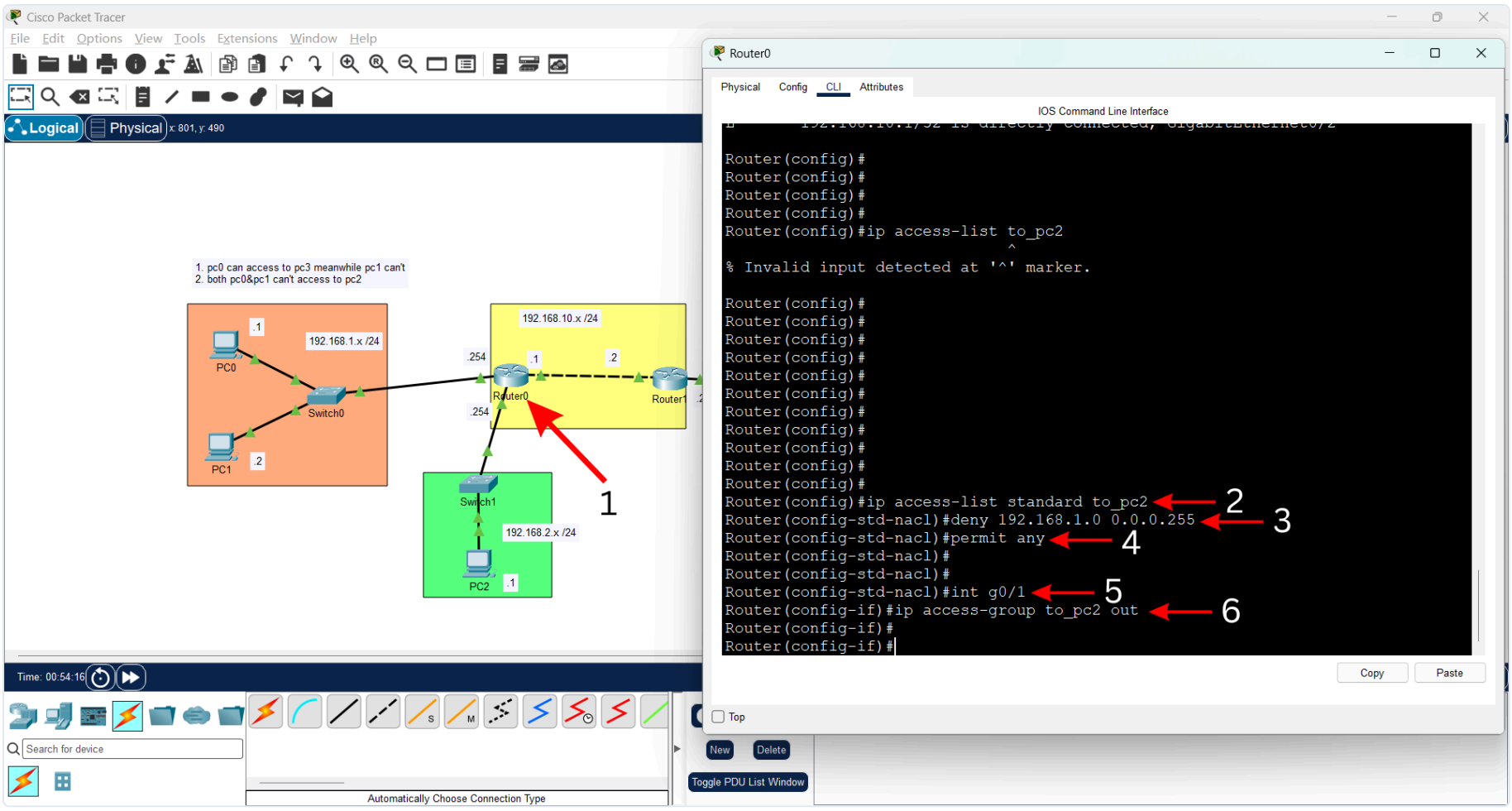
2. Policy 2 (On Router0, block 192.168.1.0/24 from PC2):

```
Router0(config)# ip access-list standard to_pc2
Router0(config-std-nacl)# deny 192.168.1.0 0.0.0.255
Router0(config-std-nacl)# permit any
Router0(config-std-nacl)# exit
Router0(config)# interface gigabitethernet 0/1
Router0(config-if)# ip access-group to_pc2 out
Router0(config-if)# exit
```

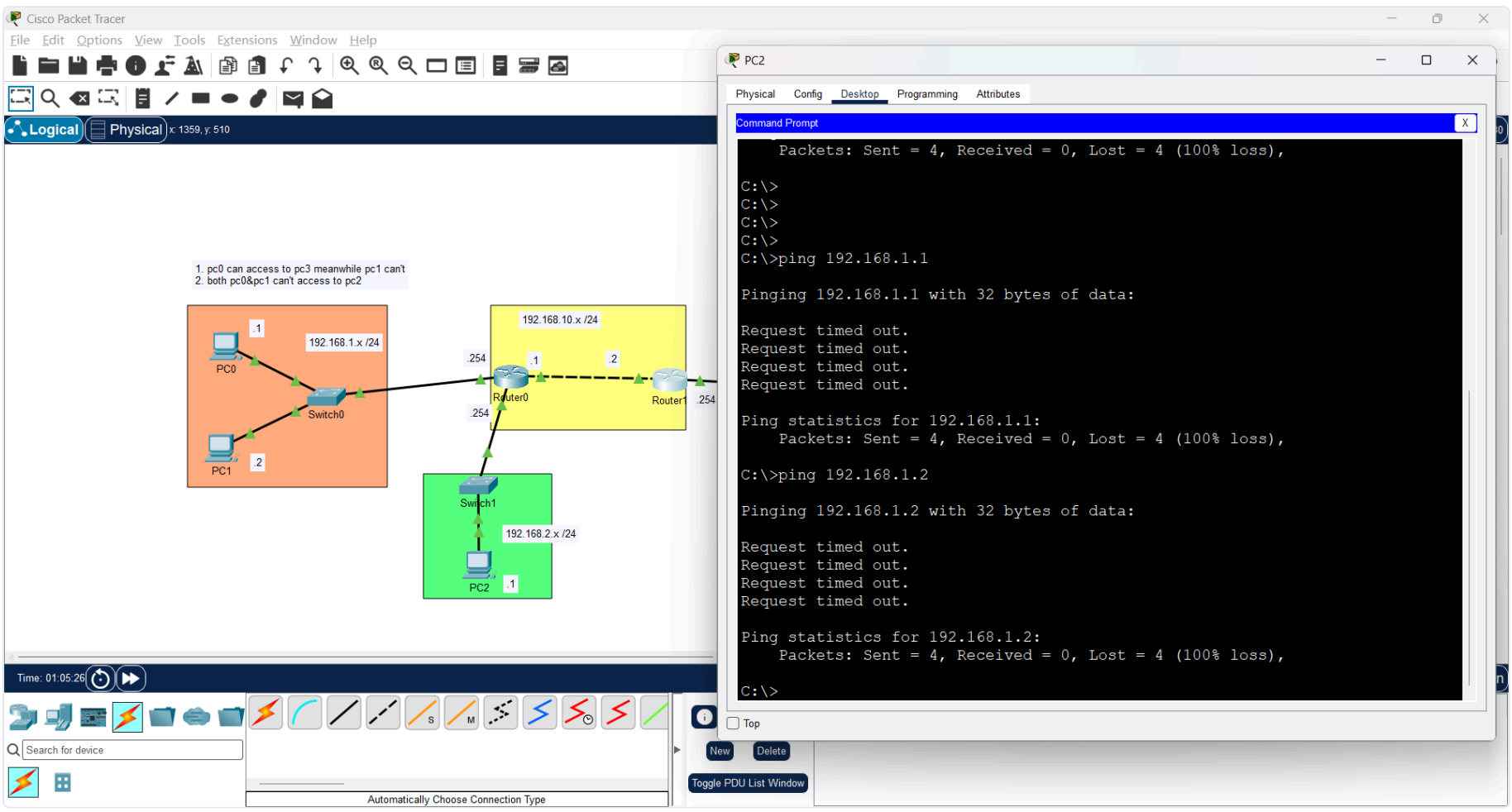


Step 5: Verification of Security Enforcement

- From PC3: Ping PC0 (`192.168.1.1`) → Success (0% loss); Ping PC1 (`192.168.1.2`) → Request timed out (Blocked).



- From PC2: Ping PC0 (`192.168.1.1`) → Request timed out (Blocked); Ping PC1 (`192.168.1.2`) → Request timed out (Blocked).



4. Lab 4 Question Solutions (Grading Guide)

Question 1: Standard ACL Placement Rule

Why are standard ACLs typically placed as close to the destination as possible (e.g., outbound on the target LAN interface) rather than near the source?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Filtering Limitation (2 Marks):* Standard ACLs inspect only the source IP address and cannot specify destination addresses or transport protocols.
> 2. *Overblocking Prevention (3 Marks):* If a standard ACL were placed near the source (e.g., inbound on Router0 G0/0), it would block matching hosts from communicating with all network destinations. Placing it outbound on the destination LAN interface ensures that packets are filtered only when attempting to reach that specific destination.

Question 2: Implicit Deny Behavior

At the end of every Cisco Access Control List, there is an invisible implicit rule. What is this default rule, and why was `permit any` required in ACL `to_pc2` ?

> **[Model Answer & Scoring Criteria - 5 Marks]:**
> 1. *Implicit Deny Definition (2 Marks):* Every Cisco ACL concludes with an invisible default rule: `deny any` (implicit deny). Any packet that does not match an explicit permit statement is discarded.
> 2. *Permit Any Necessity (3 Marks):* In ACL `to_pc2` , the goal was to deny only subnet `192.168.1.0/24` . Without adding `permit any` , legitimate traffic from other networks (such as PC3 at `192.168.3.1`) would match the implicit deny and be unintentionally blocked.

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5. Comprehensive Workshop Assessment Rubric

Evaluation Domain	Exemplary (Full Marks)	Competent (Partial Marks)	Developing (Needs Review)	Weight
Lab 1: L2 Addressing	Accurately cables topology; explains subnet boundary failure; correctly reconfigures PC1 to restore connectivity.	Cabling and IP assignment complete; minor uncertainty explaining why cross-subnet ping failed.	Unable to configure host IPs; persistent failure to achieve green link state.	20%
Lab 2: Static Routing	Configures all 6 router interfaces; brings links up with `no shut` ; deploys bidirectional static routes.	Interfaces configured correctly, but required guidance on intermediate router next-hop IP calculation.	Mismatched WAN subnet IPs; omission of return routes leaving ping unsuccessful.	30%
Lab 3: VLANs & Trunking	Creates VLAN 10/20; assigns access ports; successfully brings up 802.1Q trunk link and verifies cross-VLAN isolation.	Access ports configured; required assistance enabling trunk mode on inter-switch link.	Ports left in default VLAN 1; ping succeeds between unintended hosts.	25%
Lab 4: Standard ACLs	Establishes routed baseline; applies host-specific and wildcard subnet ACLs; verifies correct permit/deny enforcement.	ACL written correctly but applied to wrong interface or wrong direction (`in` vs `out`).	Omitted `permit any` , causing unintended total network lockout.	25%

6. Instructor Diagnostic Playbook

Reported Issue	Underlying Root Cause	Diagnostic & Remediation Sequence
Router link LED remains Red	Physical interface is administratively down.	Run `show ip int bri` . If status indicates "administratively down", execute `no shutdown` under interface mode.
Destination host unreachable	1. Host Default Gateway is missing. 2. Local gateway has no route to the destination network.	1. Check host IP Config tab for gateway setting. 2. Run `show ip route` on gateway router to confirm the destination prefix exists.
Request timed out across routers	Return route missing on downstream router.	Trace routing tables across all routers along the path: confirm every router has an entry returning to the source network.
Same VLAN cannot communicate across switches	Inter-switch link is operating in Access mode or required VLANs are not allowed on the Trunk.	Run `show interfaces trunk` on both switches. Verify port mode is "on/trunking" and VLANs 10, 20 are in the allowed list.

Reported Issue	Underlying Root Cause	Diagnostic & Remediation Sequence
All hosts blocked after configuring ACL	Hit by default `deny any` rule at the end of the ACL.	Check ACL definition using `show access-lists`. Ensure `permit any` is appended when only specific subnets should be denied.